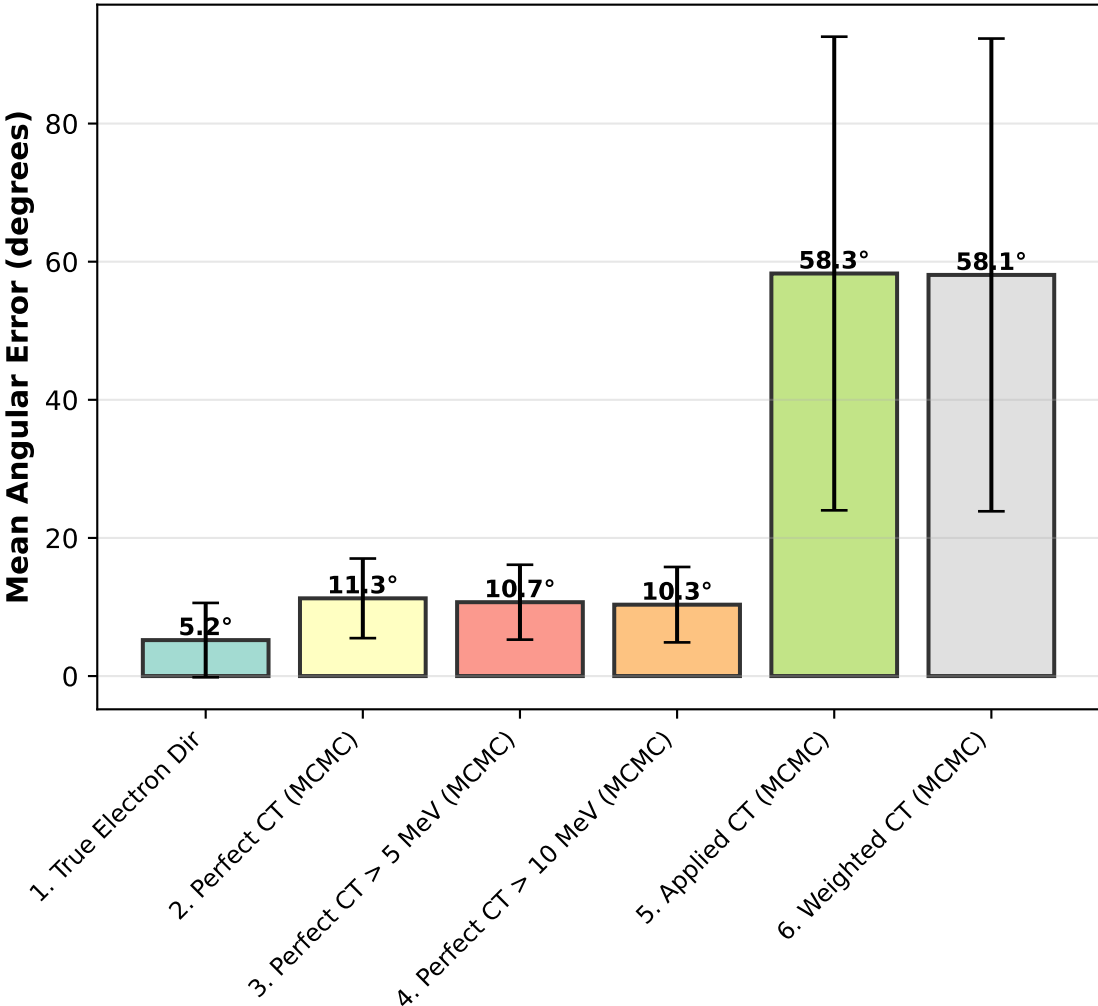


Emcee 8-Scenario Aggregate Analysis

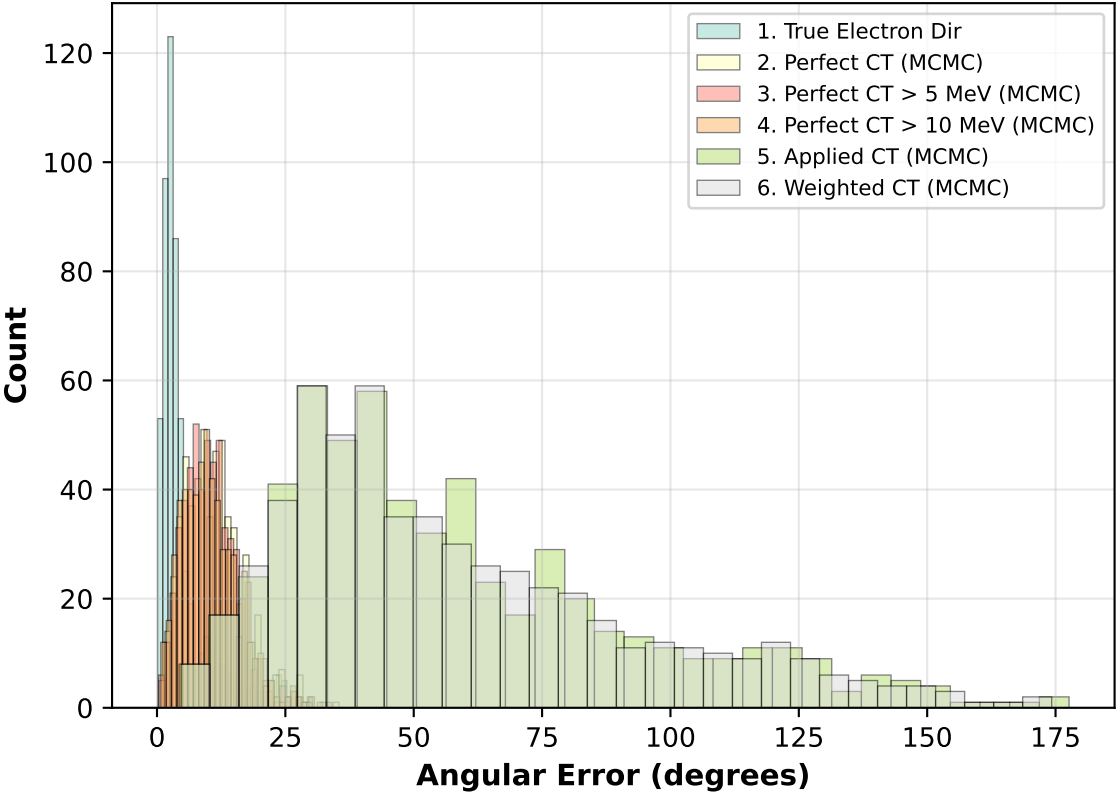
Summary Statistics

Scenario	N	68% Q	Median	Mean
1. True Electron Dir	567	4.72°	3.28°	5.21°
2. Perfect CT (MCMC)	567	13.25°	10.58°	11.26°
3. Perfect CT > 5 MeV (MCMC)	567	12.54°	10.00°	10.70°
4. Perfect CT > 10 MeV (MCMC)	567	12.17°	9.64°	10.34°
5. Applied CT (MCMC)	567	67.16°	48.68°	58.29°
6. Weighted CT (MCMC)	567	67.21°	48.86°	58.08°

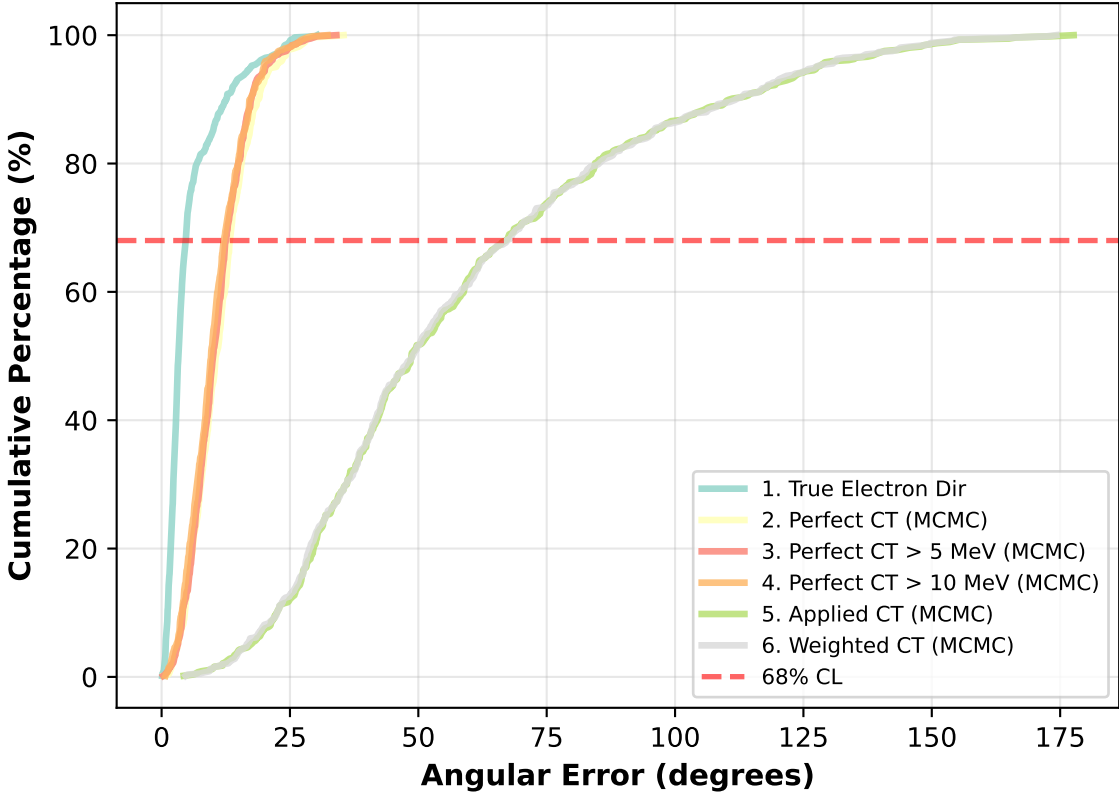
Mean Resolution ± Std Dev



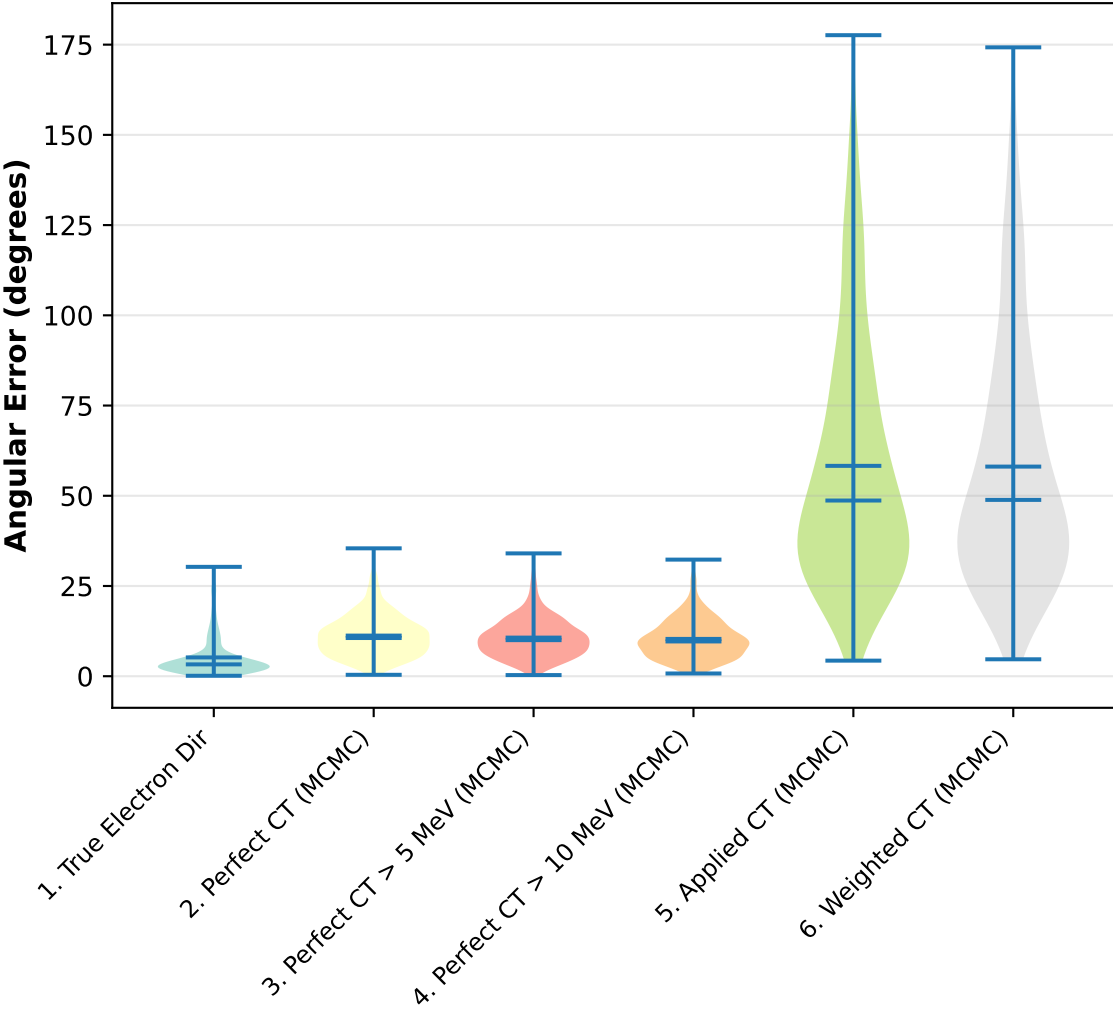
Distribution Overlay



Cumulative Distribution

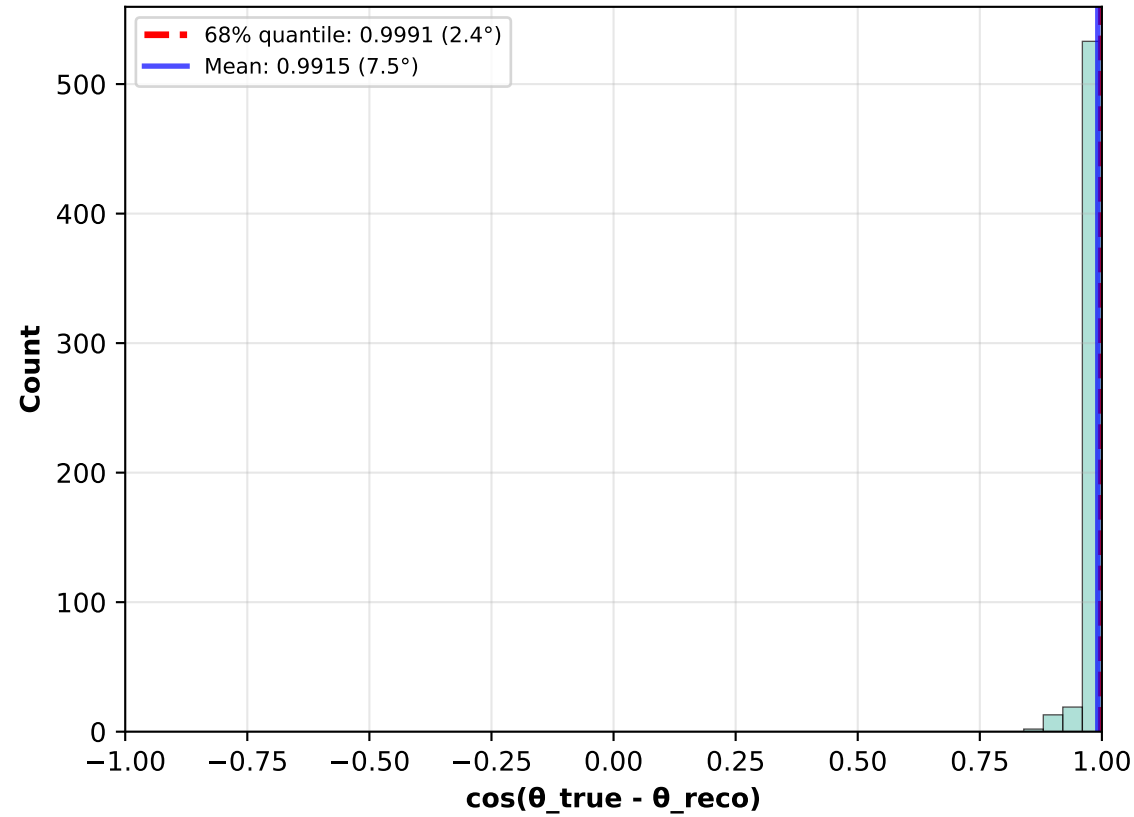


Distribution Shapes

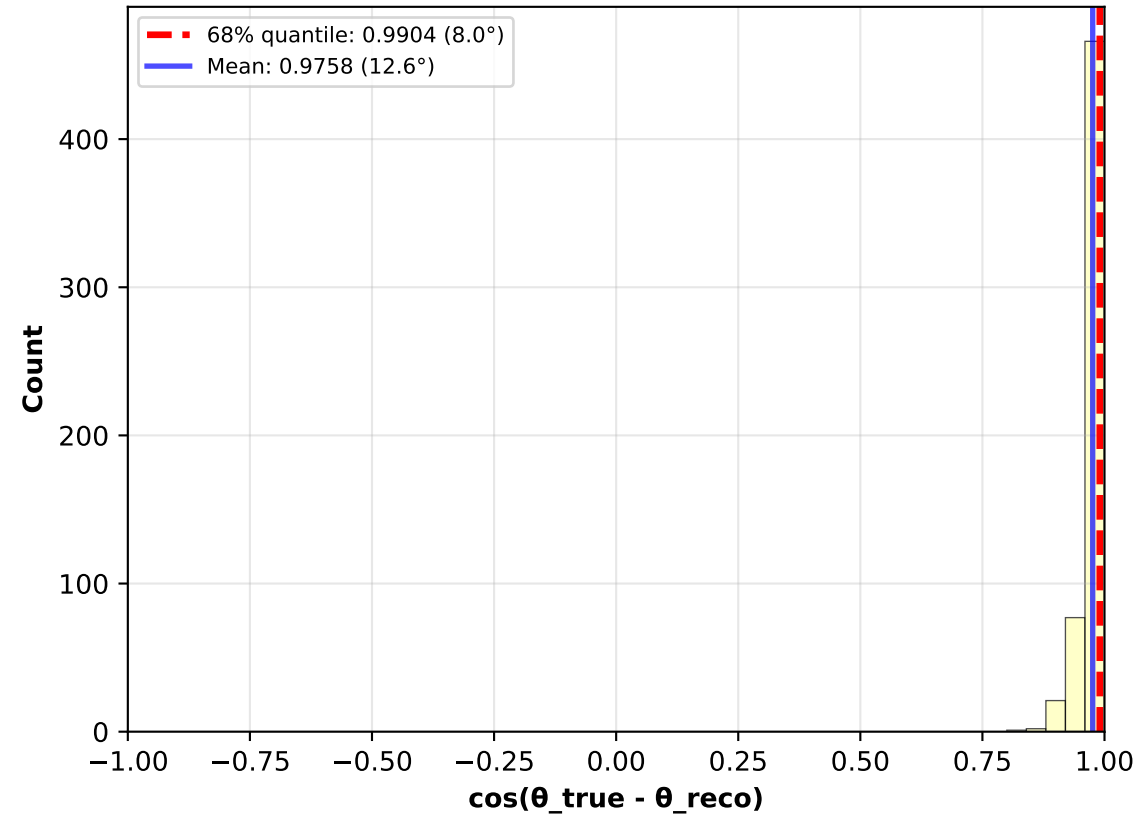


Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Full Range

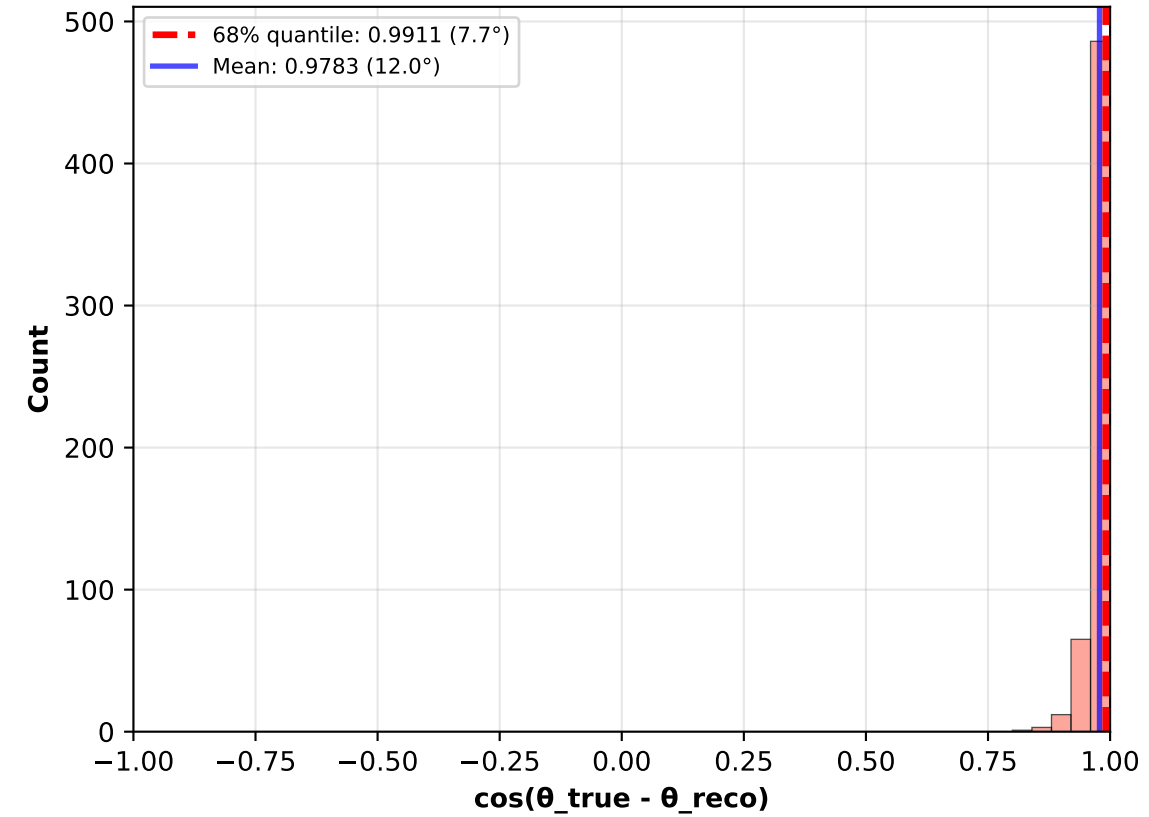
1. True Electron Dir
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)



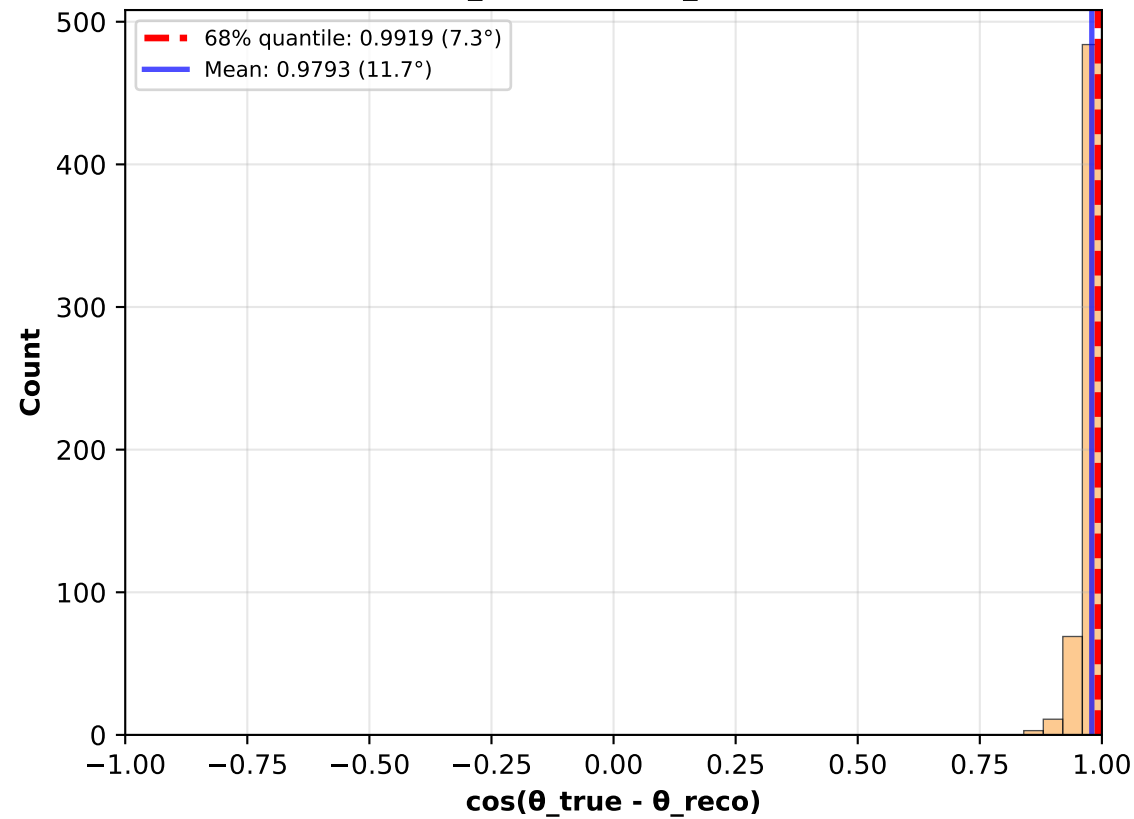
2. Perfect CT (MCMC)
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)



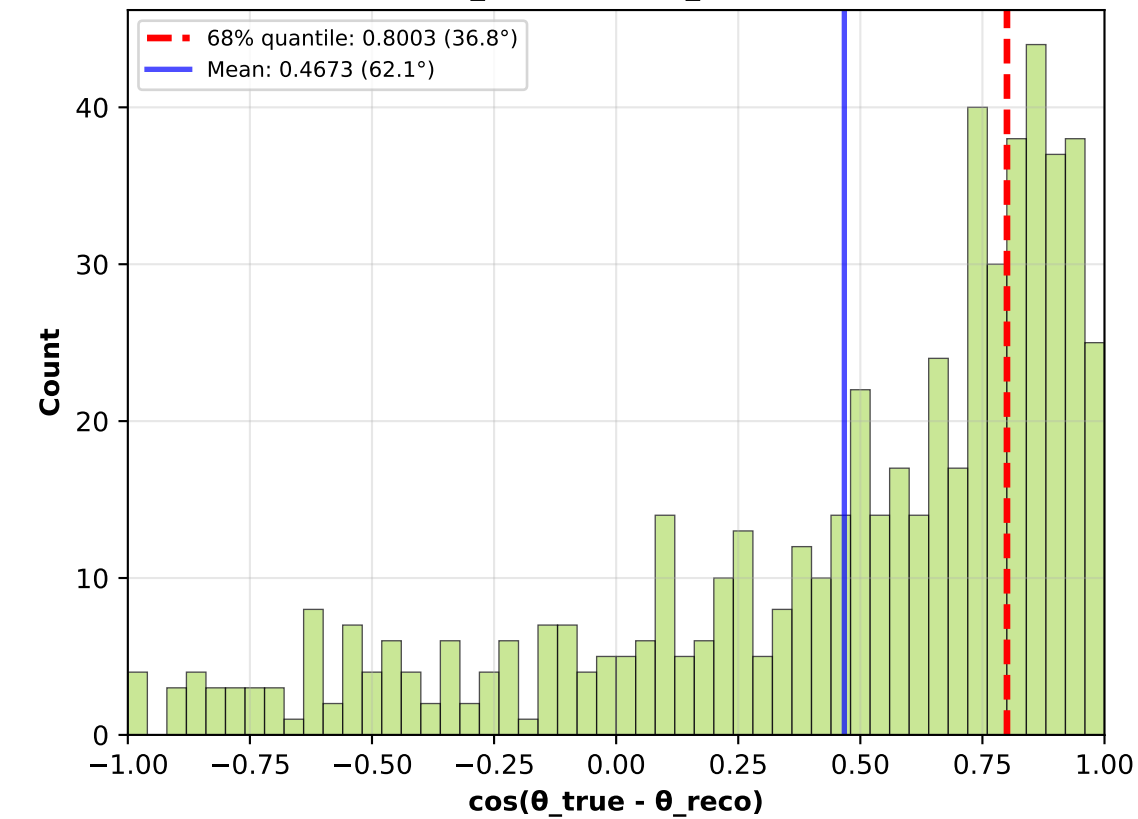
3. Perfect CT > 5 MeV (MCMC)
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)



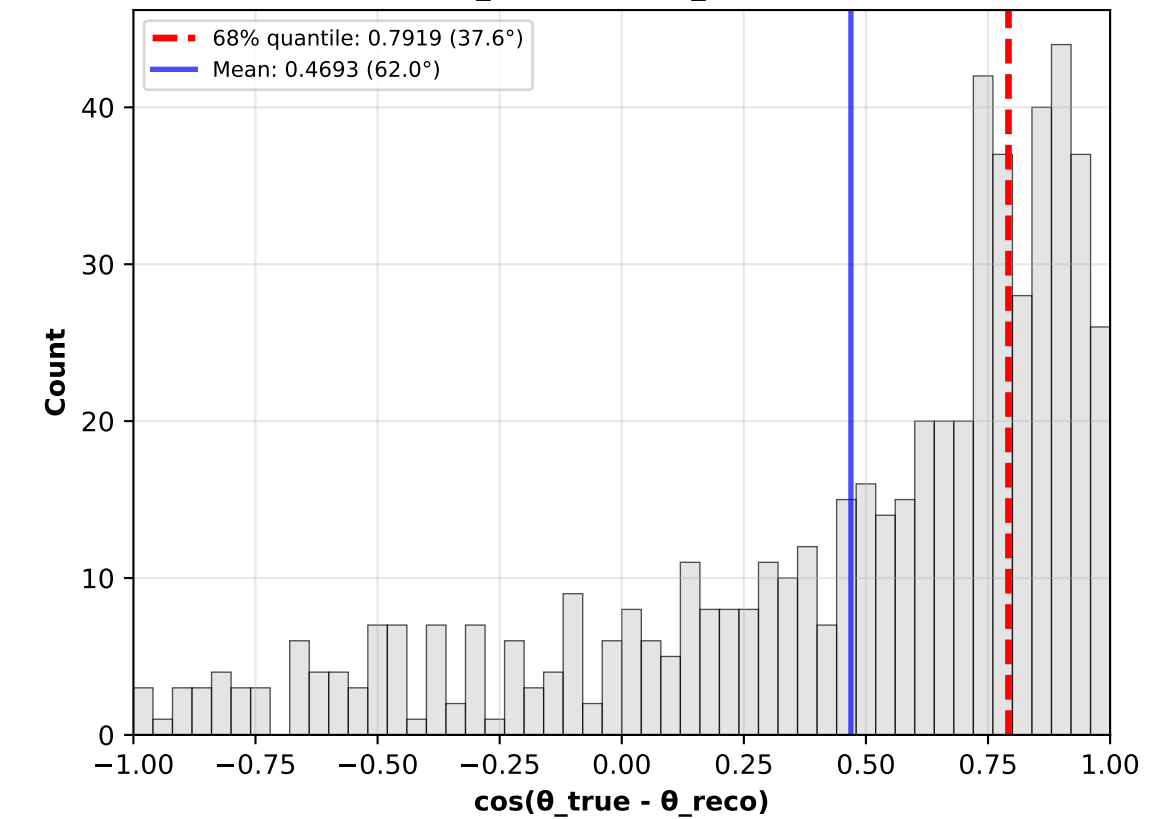
4. Perfect CT > 10 MeV (MCMC)
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)



5. Applied CT (MCMC)
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)

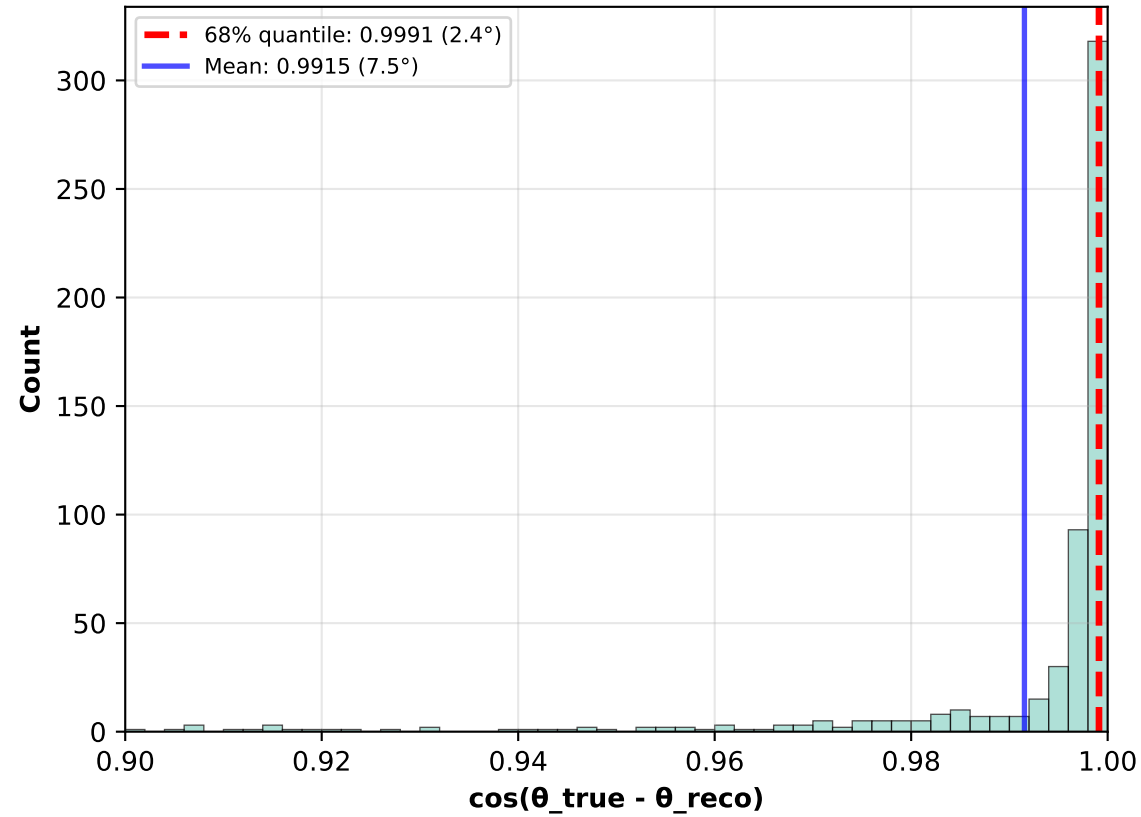


6. Weighted CT (MCMC)
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)

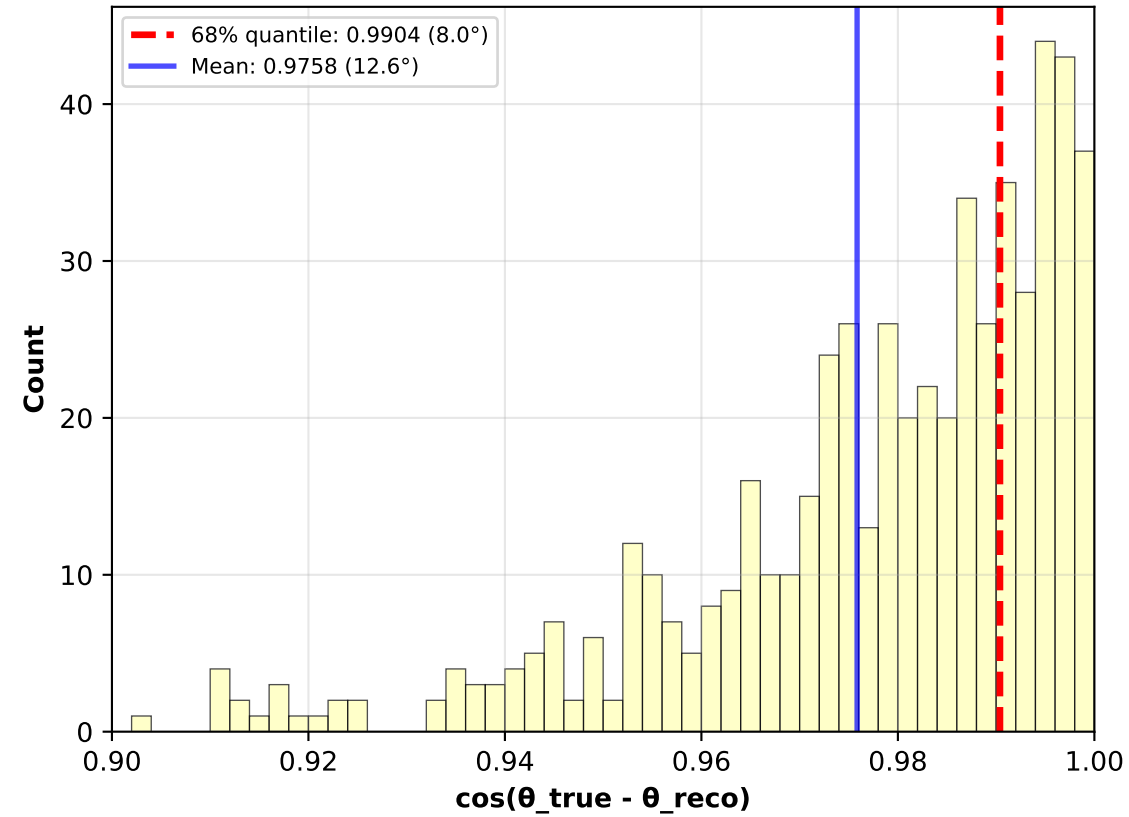


Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Zoomed (0.9-1.0)

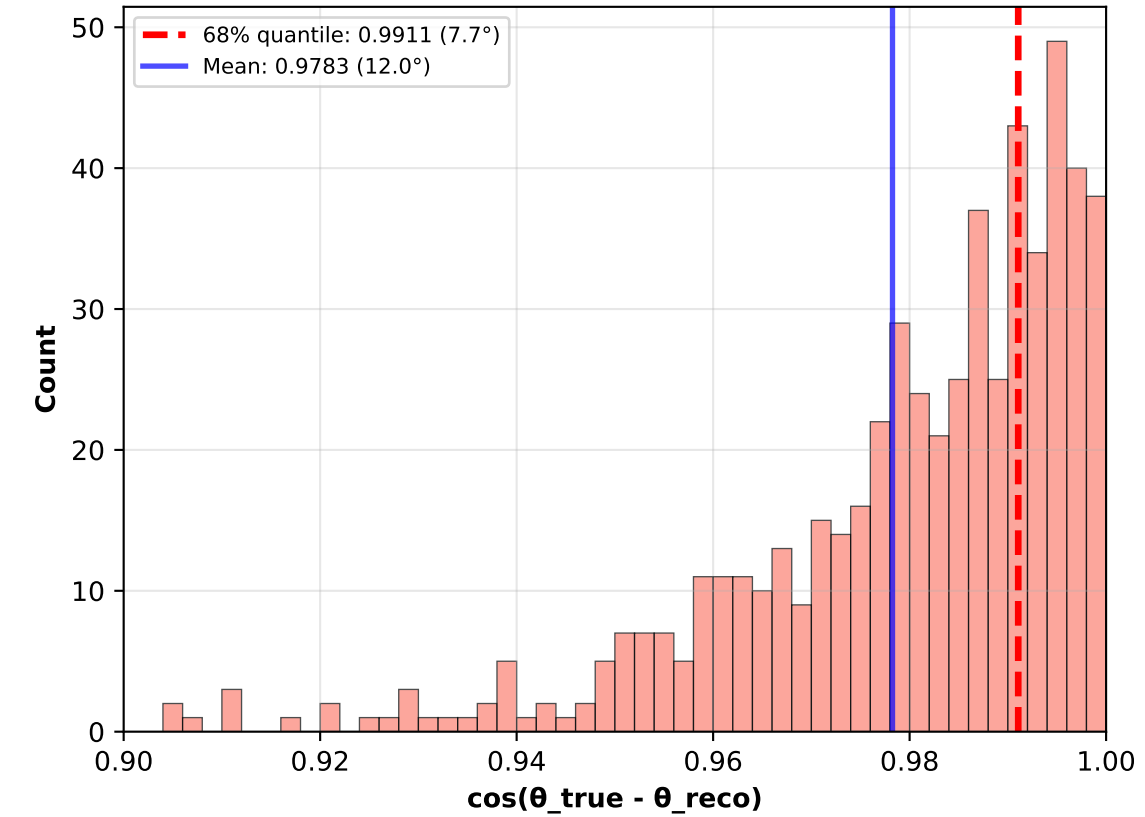
1. True Electron Dir
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)



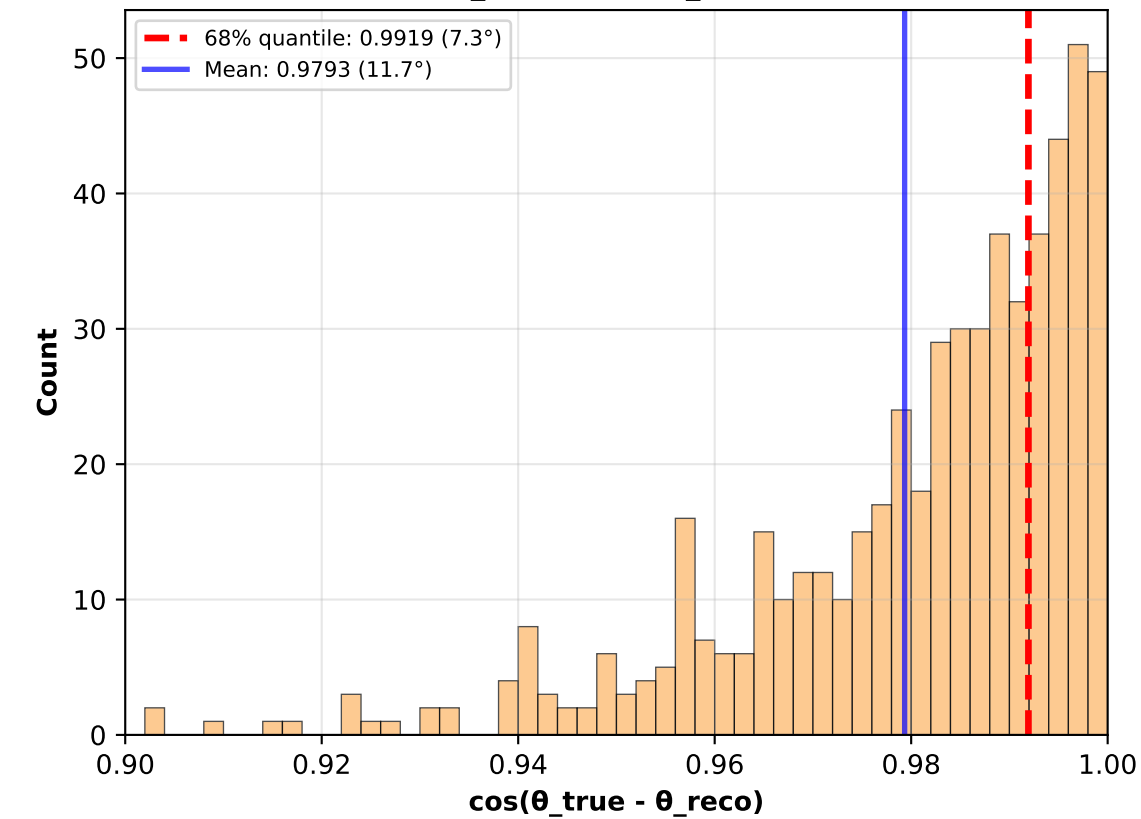
2. Perfect CT (MCMC)
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)



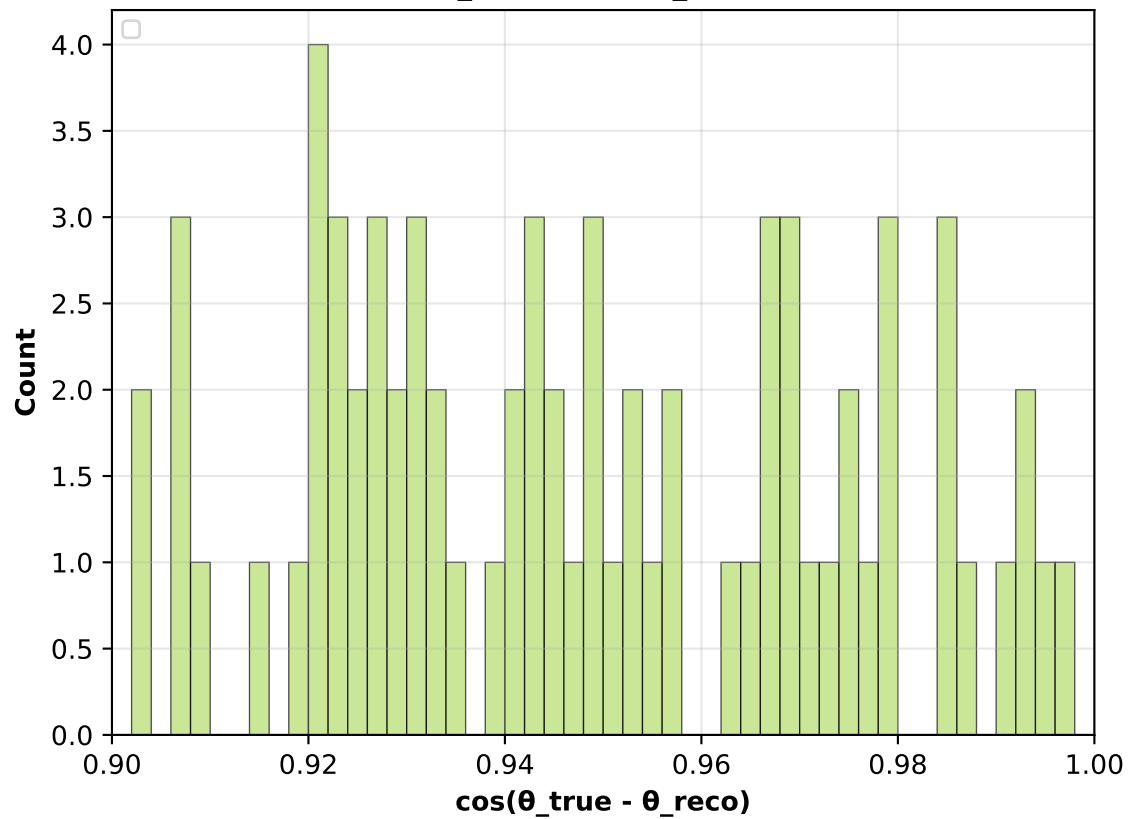
3. Perfect CT > 5 MeV (MCMC)
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)



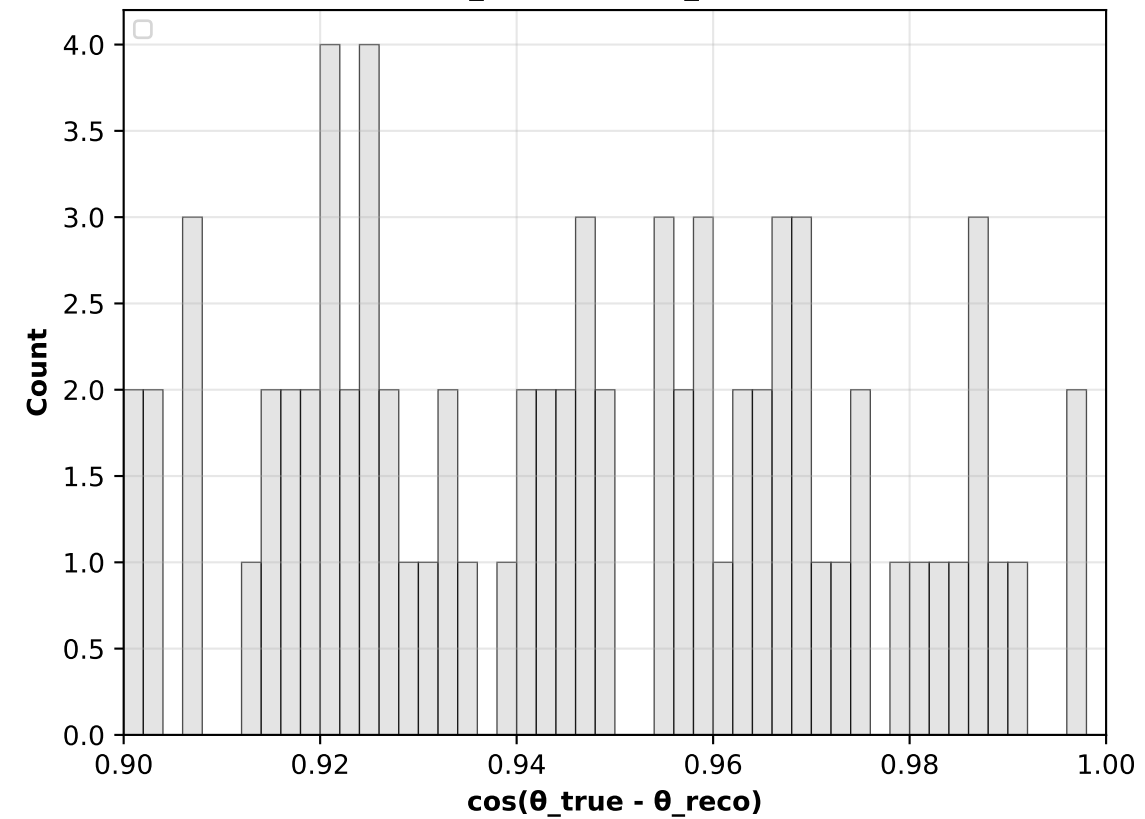
4. Perfect CT > 10 MeV (MCMC)
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)



5. Applied CT (MCMC)
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)



6. Weighted CT (MCMC)
($N_{\text{cats}}=567$, $N_{\text{ES}} \approx 0$)



Cluster Usage Across Scenarios

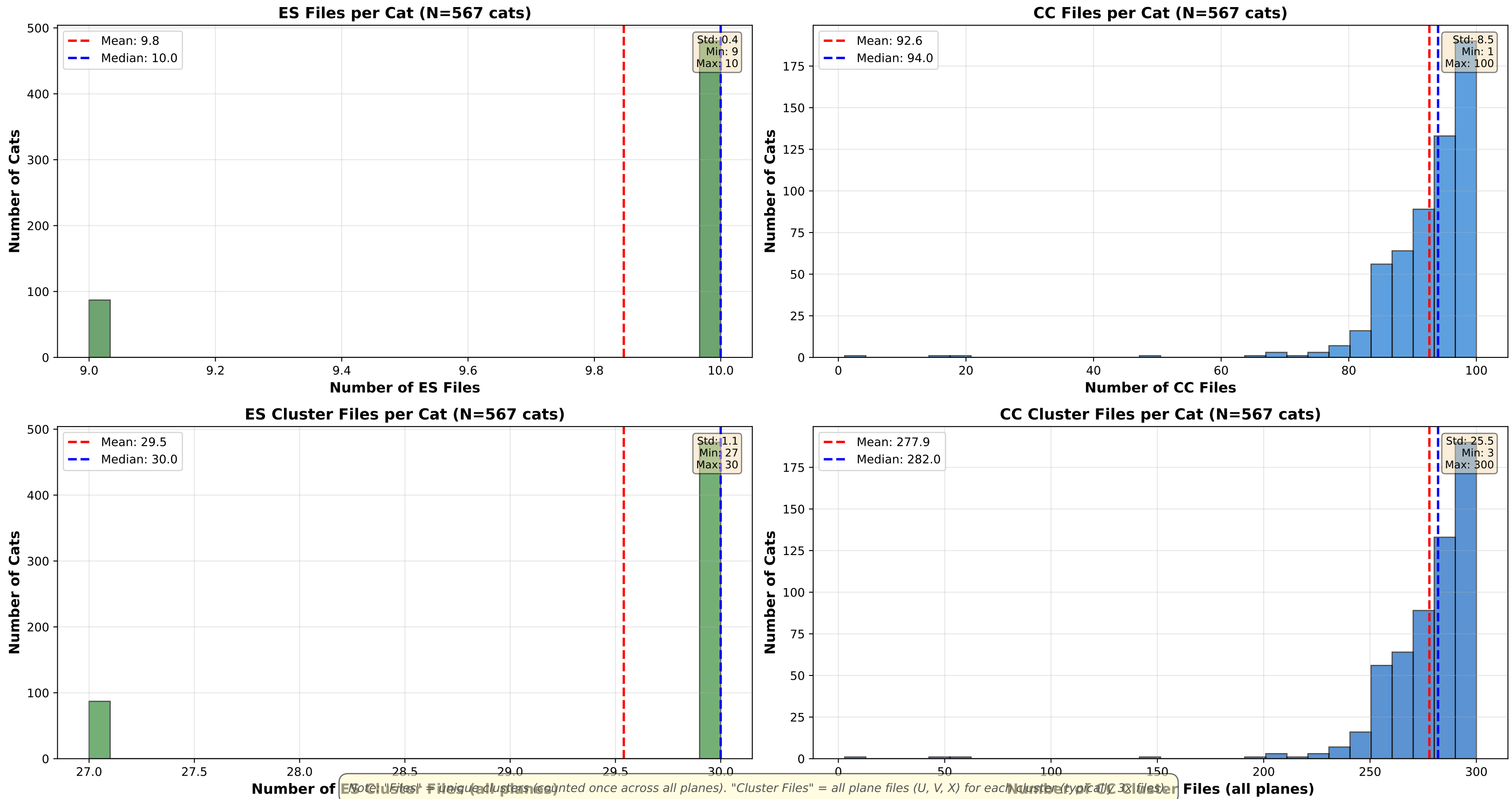
Scenario	N Cats	Input Total	Input ES Main	Value "Clusters Used"	Notes
1. True Electron Dir	567	102	10	271	MCMC samples (weighted sum)
2. Perfect CT (MCMC)	567	102	10	271	MCMC samples (weighted sum)
3. Perfect CT > 5 MeV (MCMC)	567	102	10	241	MCMC samples (weighted sum)
4. Perfect CT > 10 MeV (MCMC)	567	102	10	147	143.6% retention
5. Applied CT (MCMC)	567	102	10	3242	MCMC samples (weighted sum)
6. Weighted CT (MCMC)	567	102	10	3242	MCMC samples (weighted sum)

Note: "Clusters Used" shows different meanings per scenario:

- Best Case, Perfect CT: Number of ES main clusters actually used (after E>3 MeV cut)
- Full Pipeline, Weighted CT: Weighted sum of cluster contributions in MCMC (includes CT probabilities)
- Energy cut scenarios: Number of clusters passing energy threshold

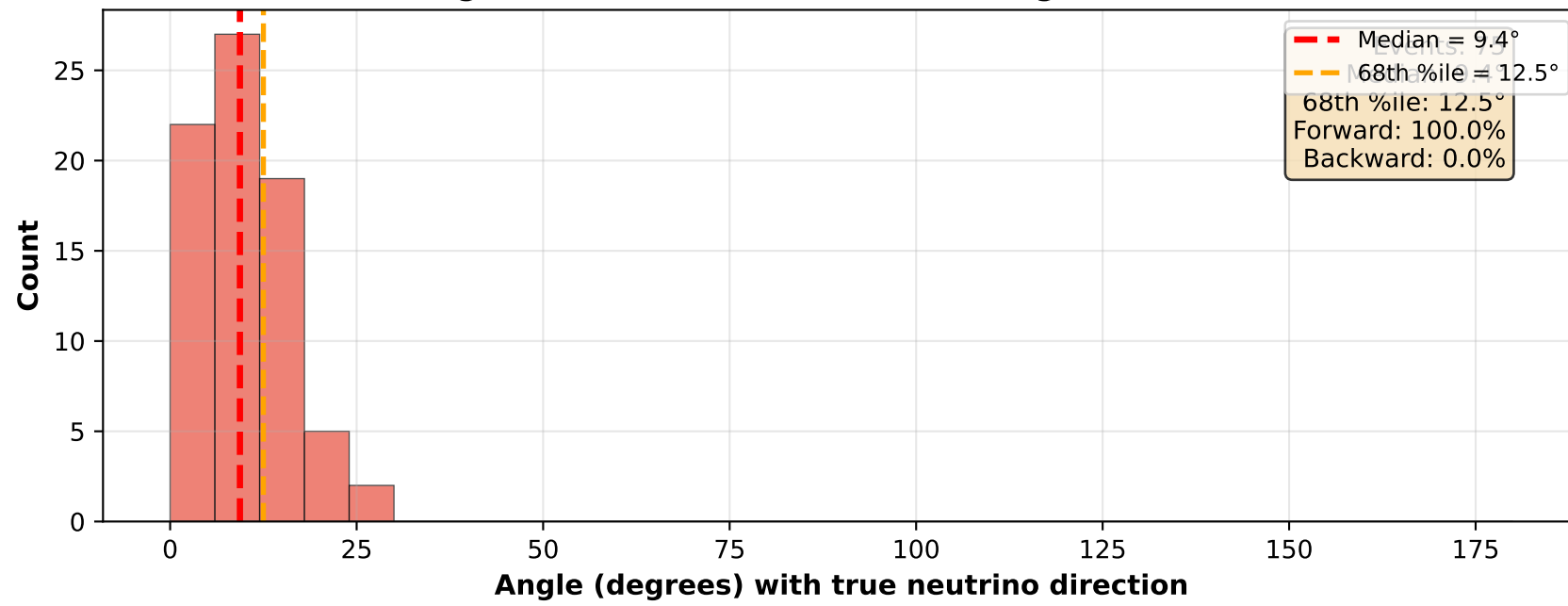
The key insight: Best Case and Perfect CT use ~85-90% of ES main tracks with E>3 MeV.
Full Pipeline shows much higher values due to weighted MCMC sampling, not actual cluster count.

Input Data Statistics: Files and Clusters per Cat

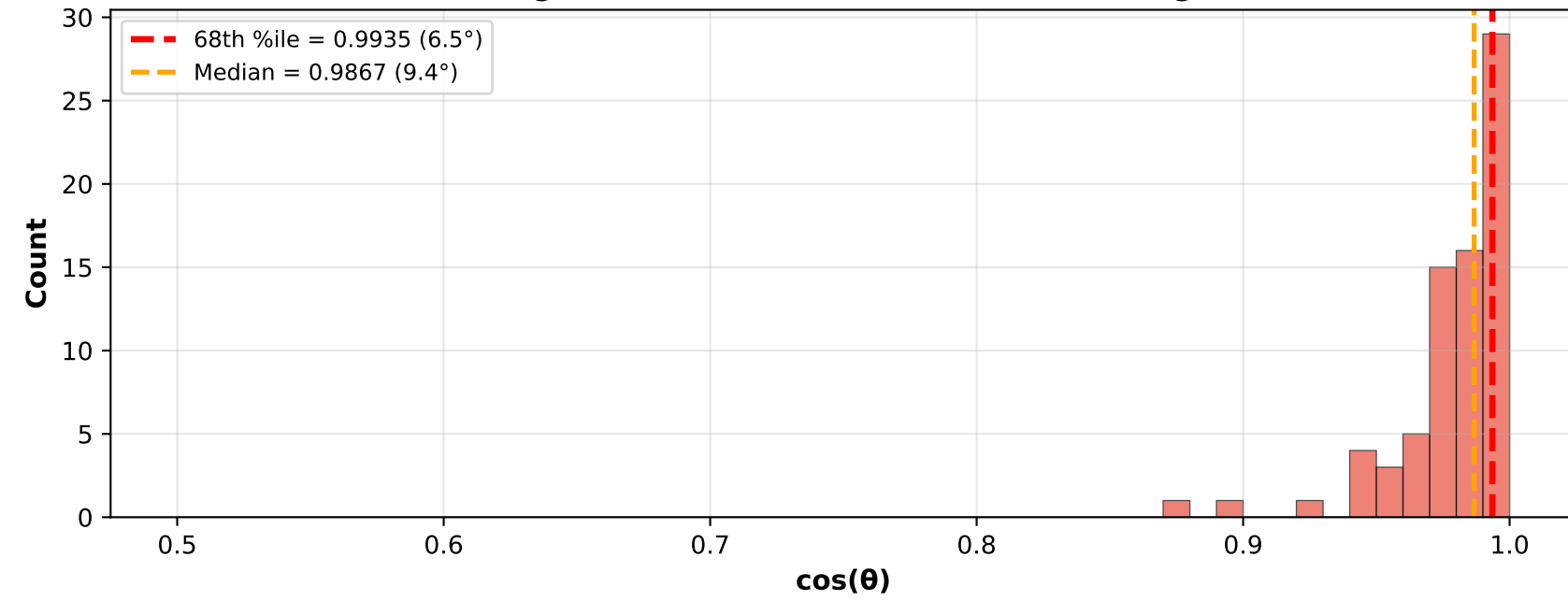


Perfect CT (ES + ED) - Axis-Aligned Analysis ($\pm 30.0^\circ$ selection)

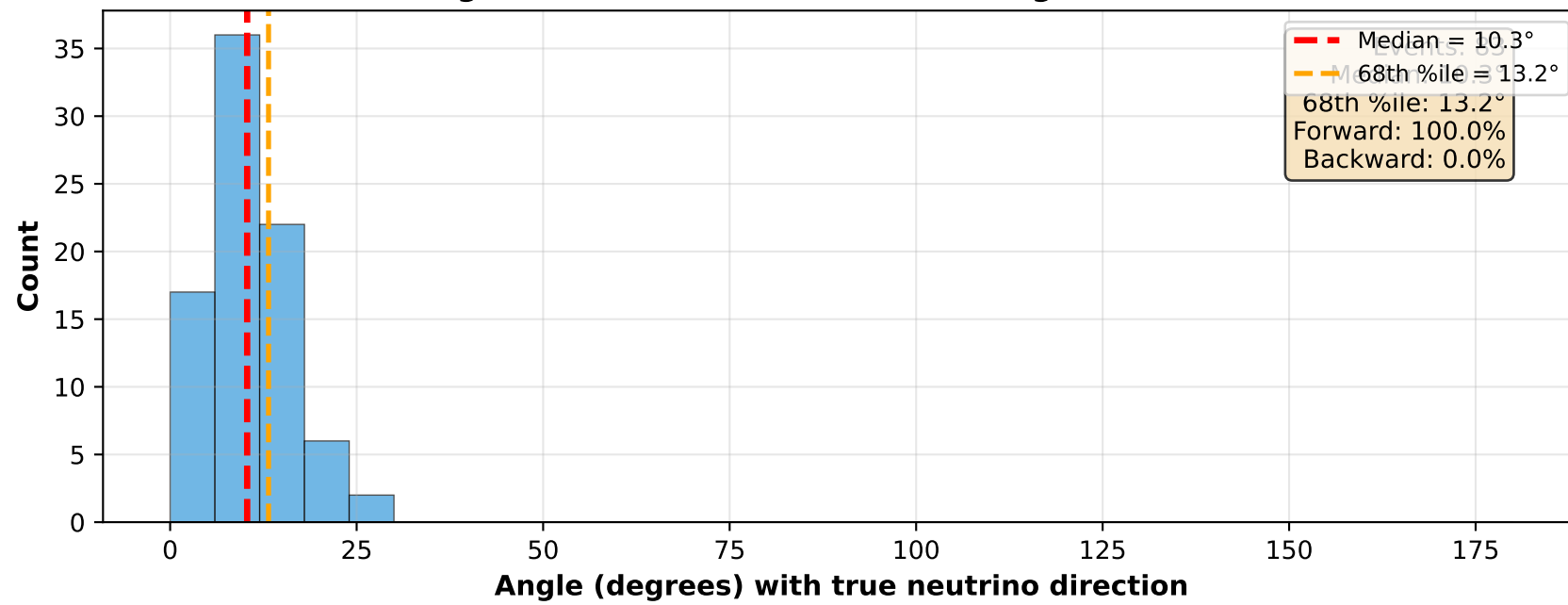
X-aligned (within $\pm 30.0^\circ$ of X-axis): Angle Distribution



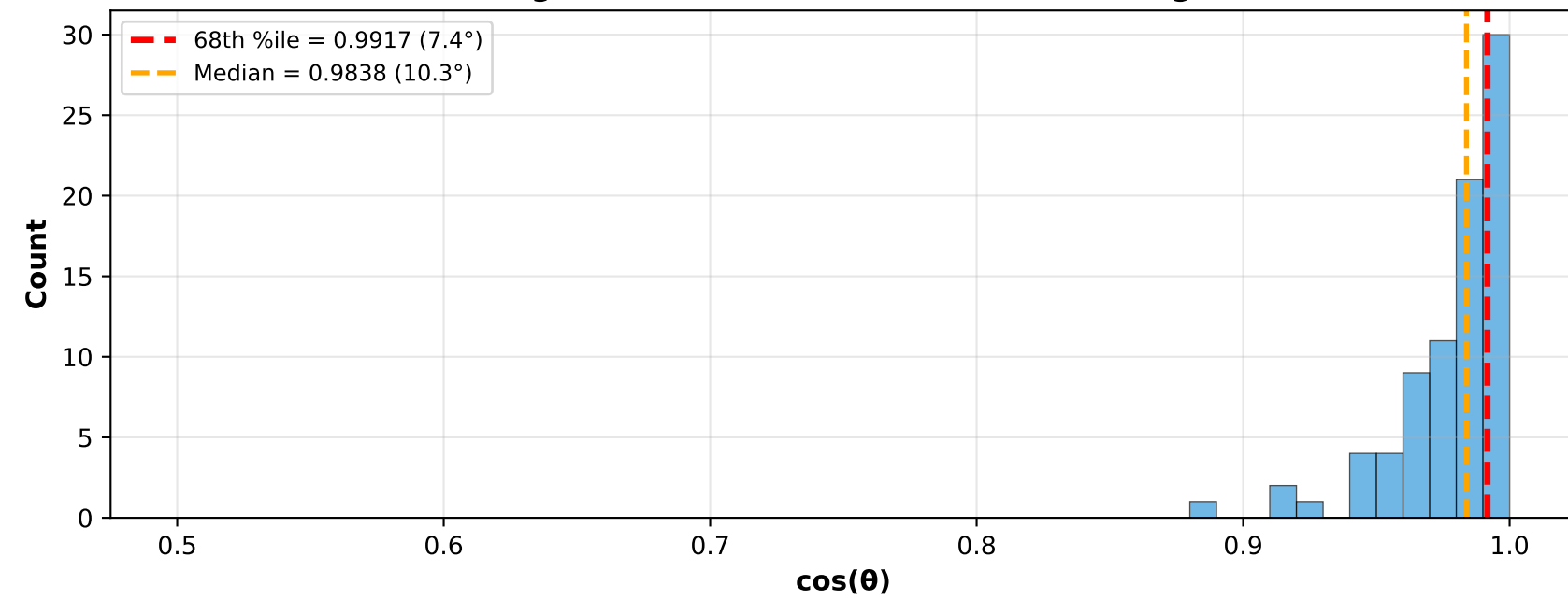
X-aligned: Cosine Distribution [0.5-1.0 range]



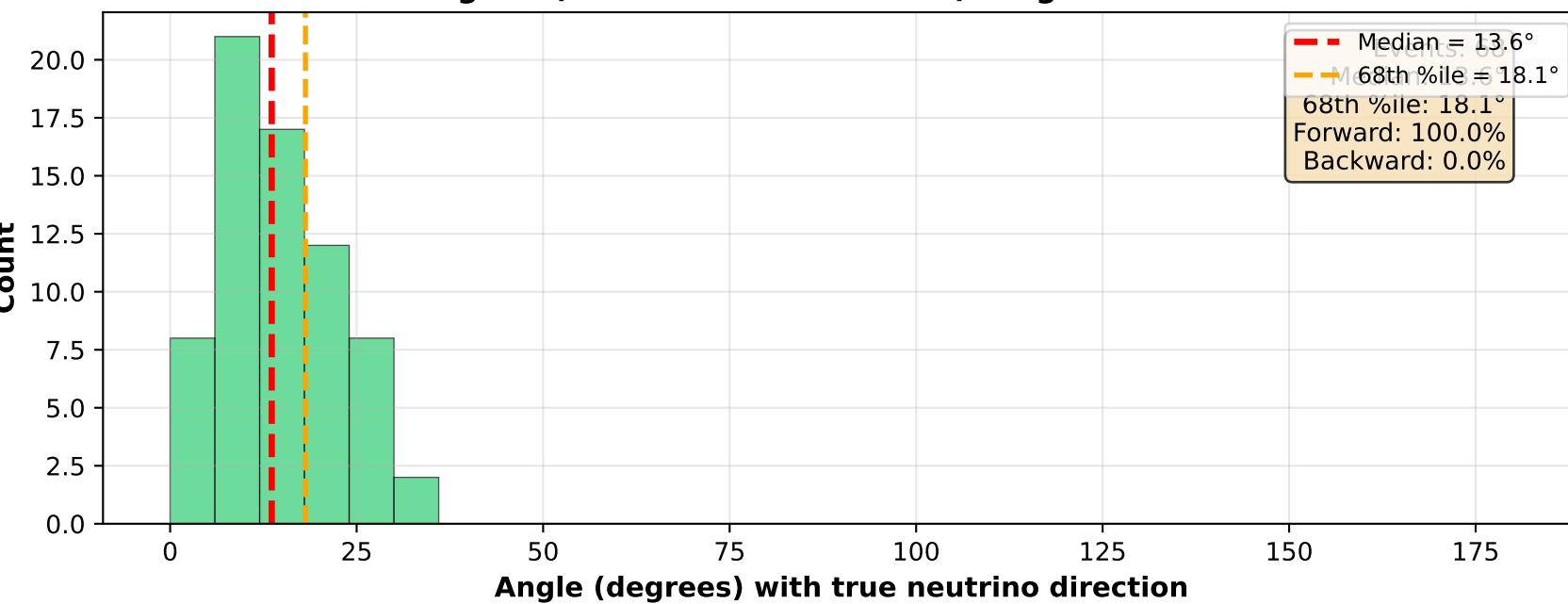
Y-aligned (within $\pm 30.0^\circ$ of Y-axis): Angle Distribution



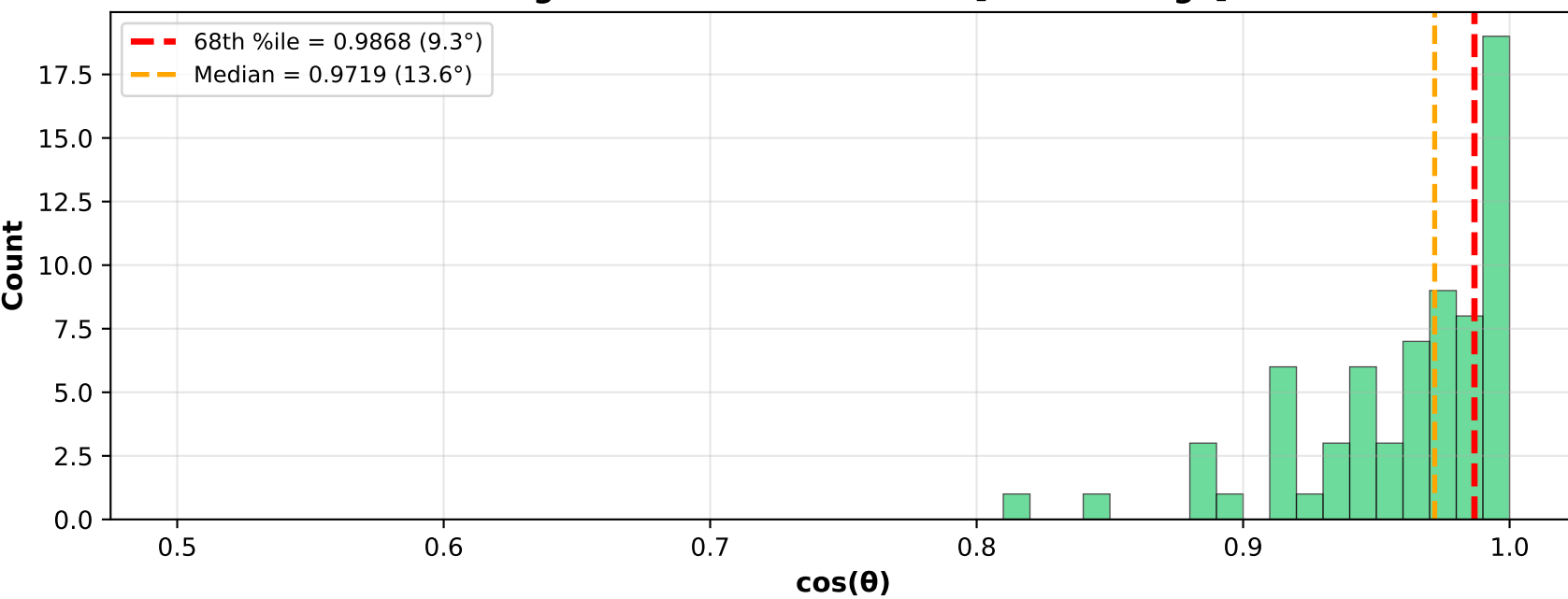
Y-aligned: Cosine Distribution [0.5-1.0 range]



Z-aligned (within $\pm 30.0^\circ$ of Z-axis): Angle Distribution



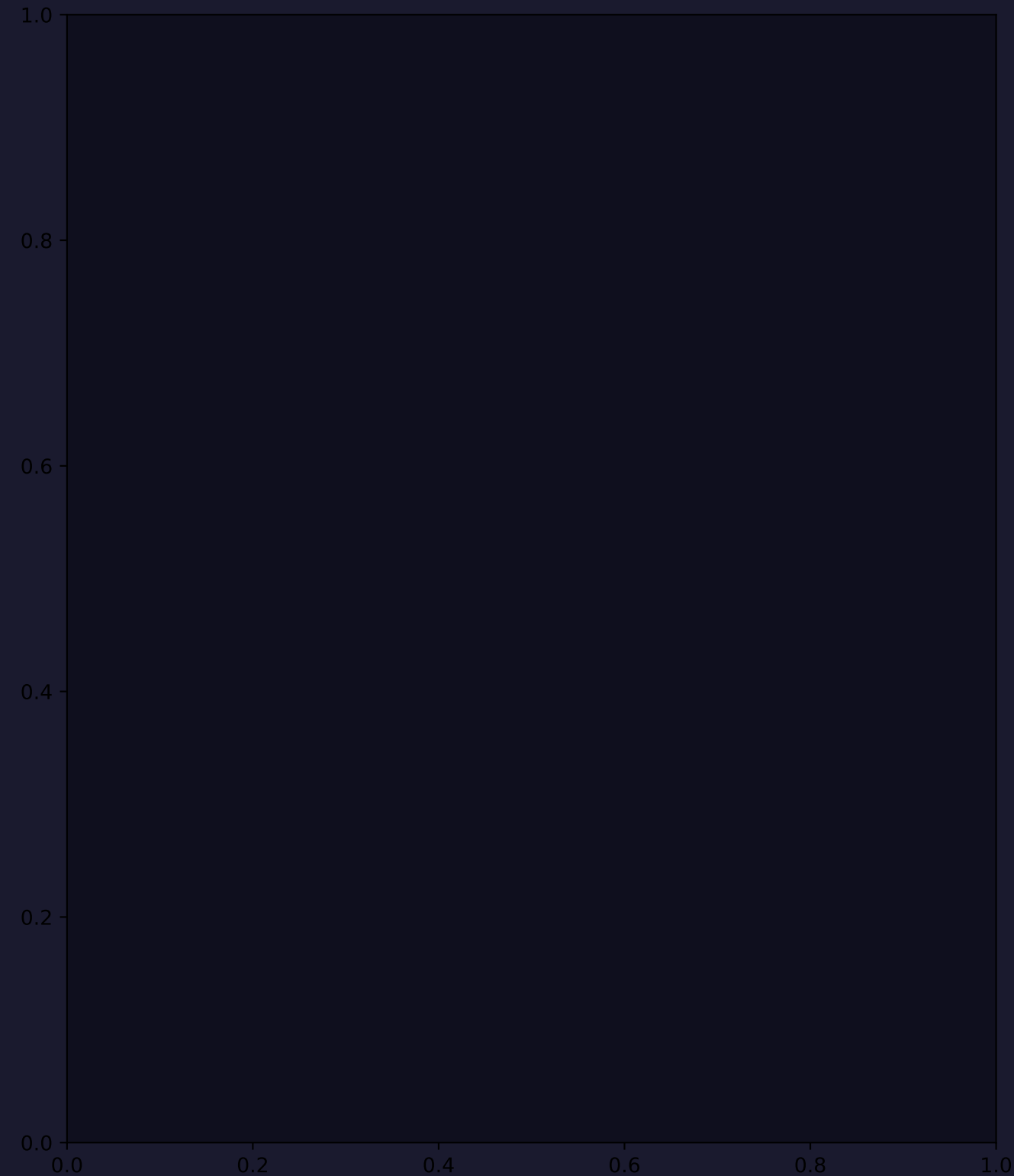
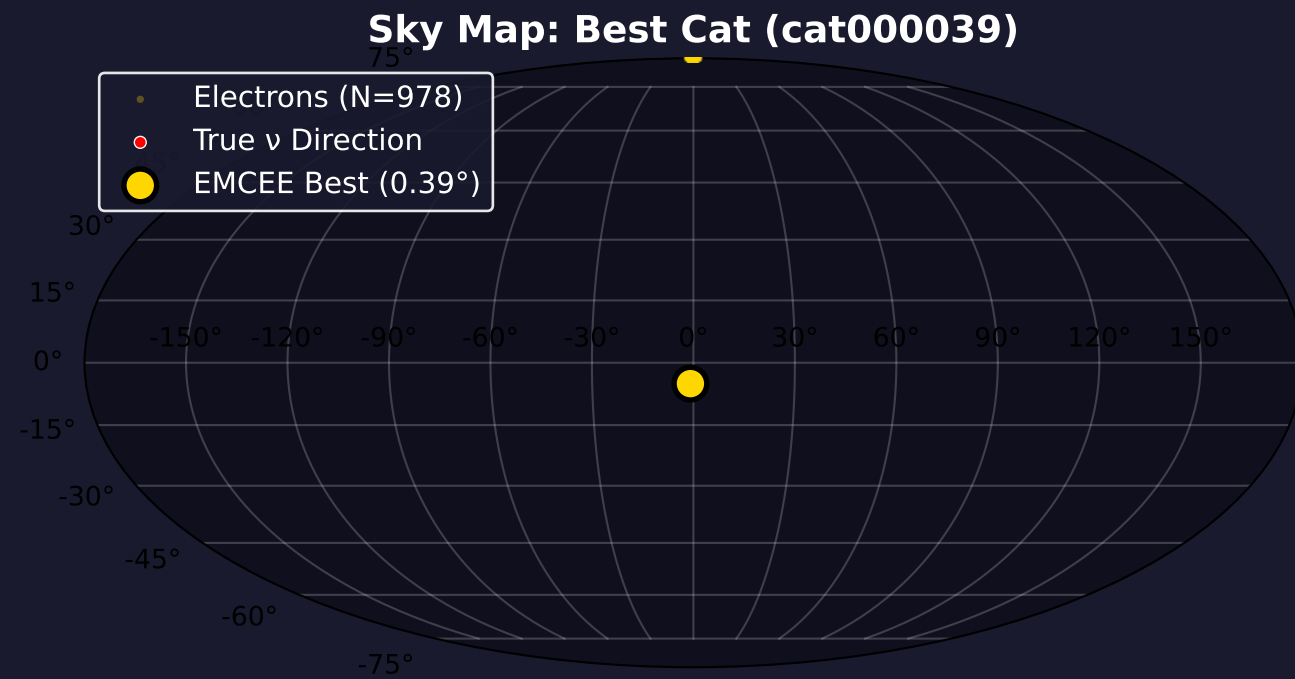
Z-aligned: Cosine Distribution [0.5-1.0 range]



NOTE: Axis alignment is based on TRUE neutrino direction. X-axis performance depends on detector geometry (drift direction, wire orientation). Verify X/Y/Z coordinate definitions match your detector setup.

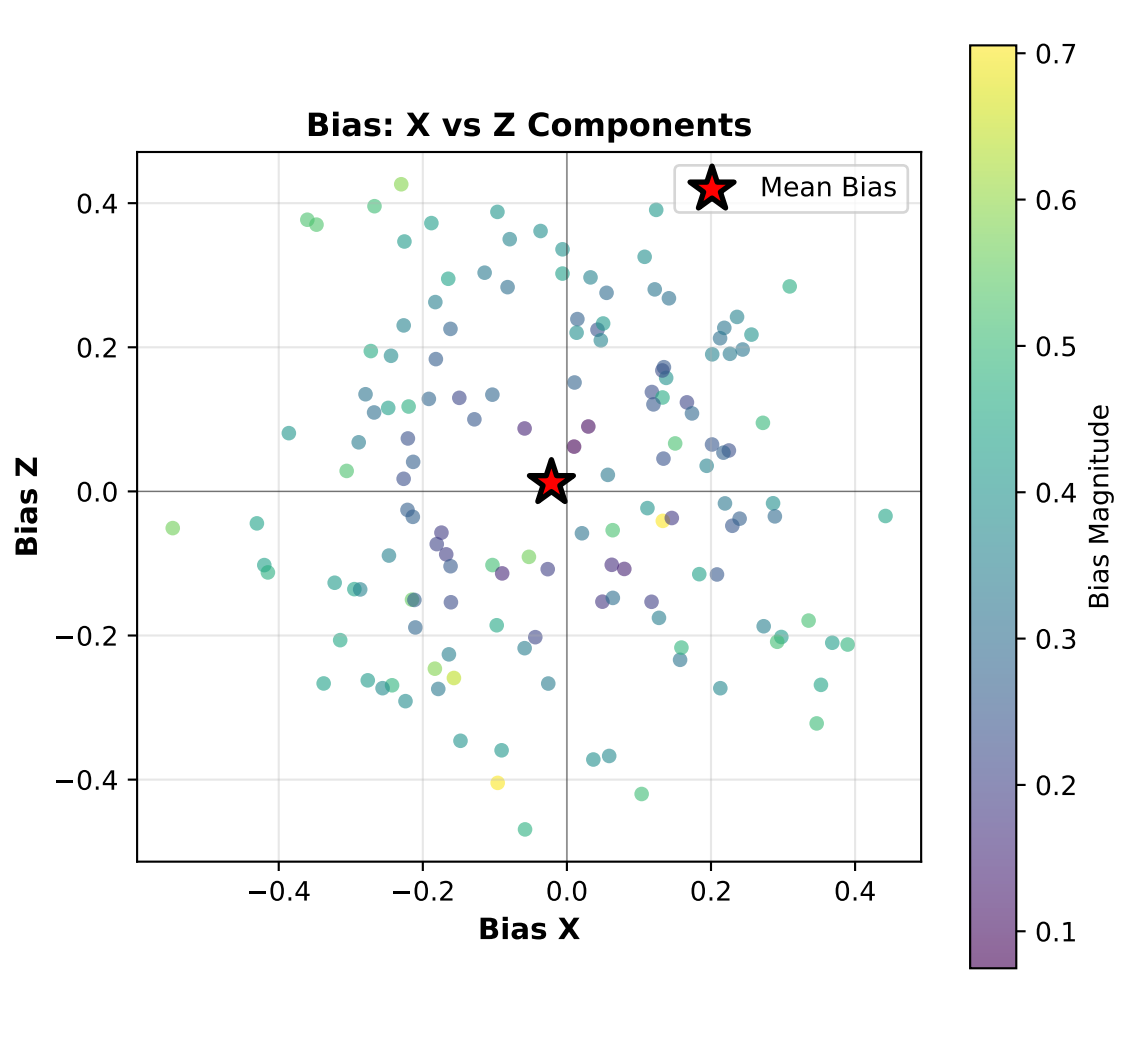
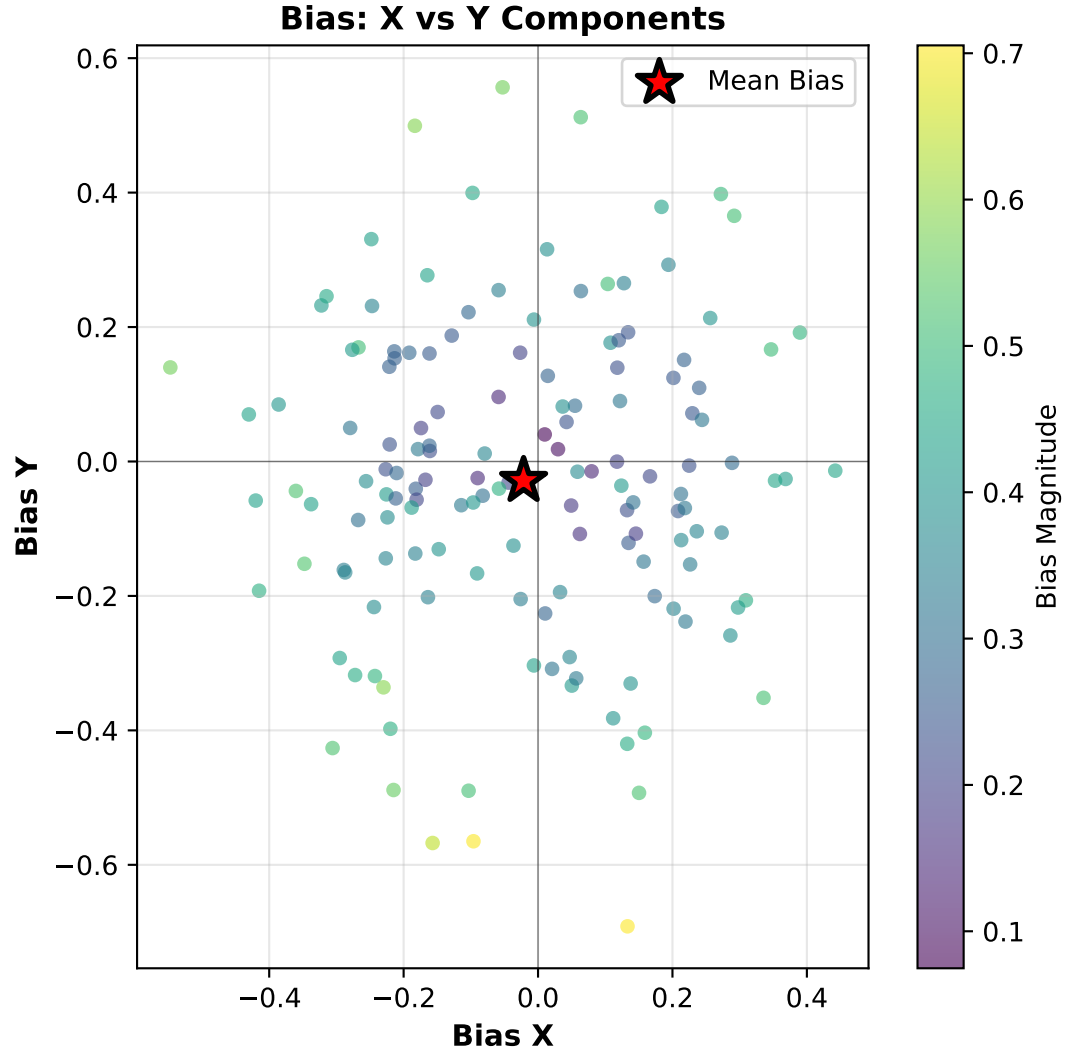
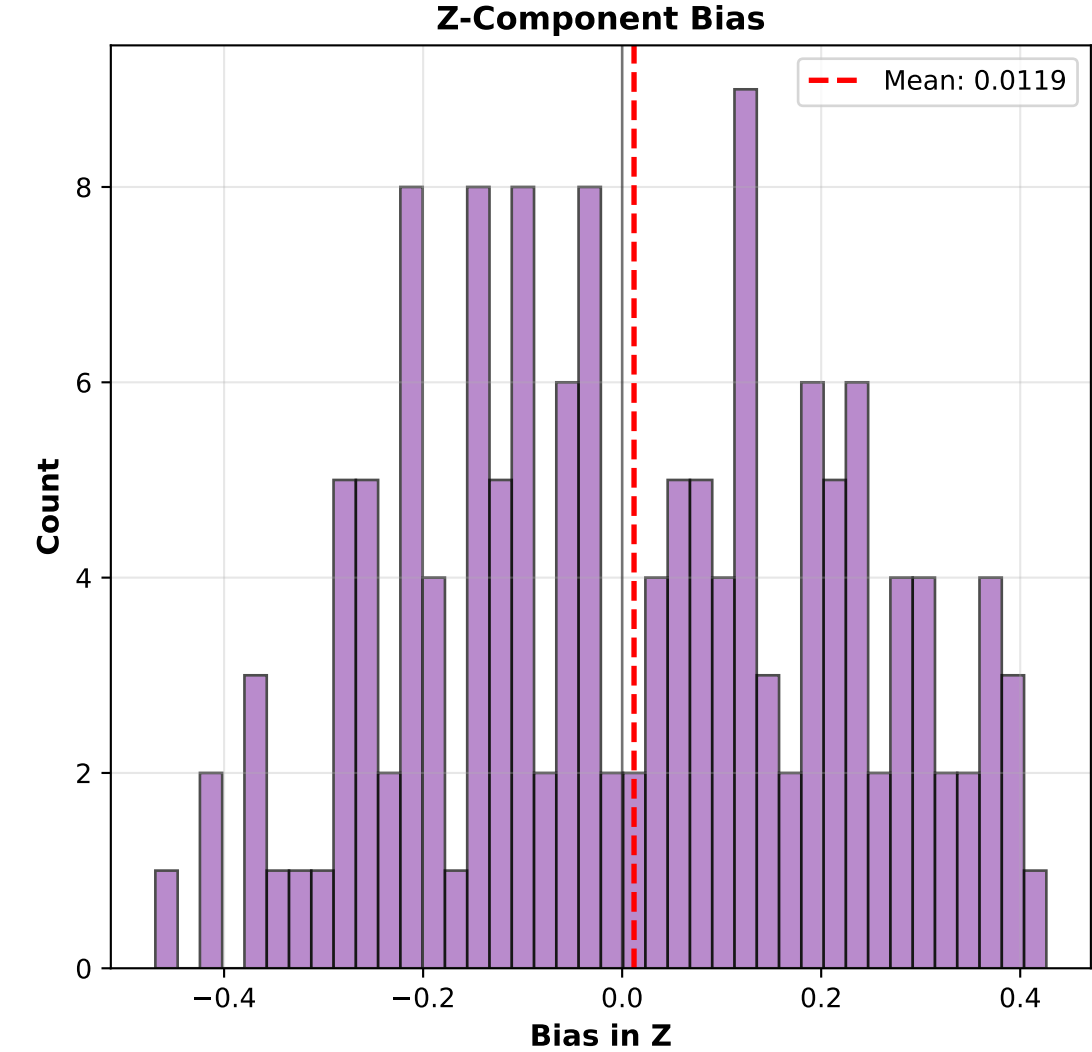
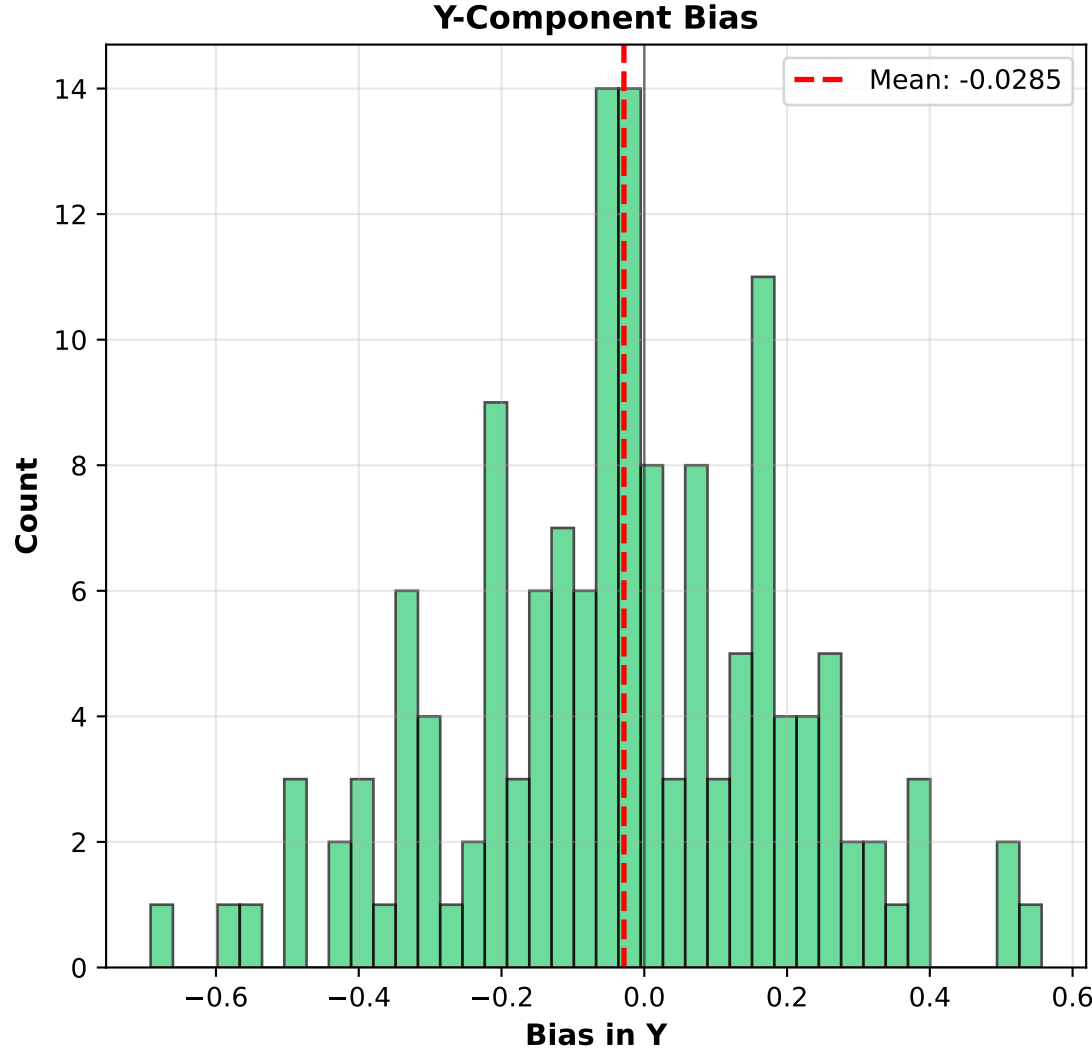
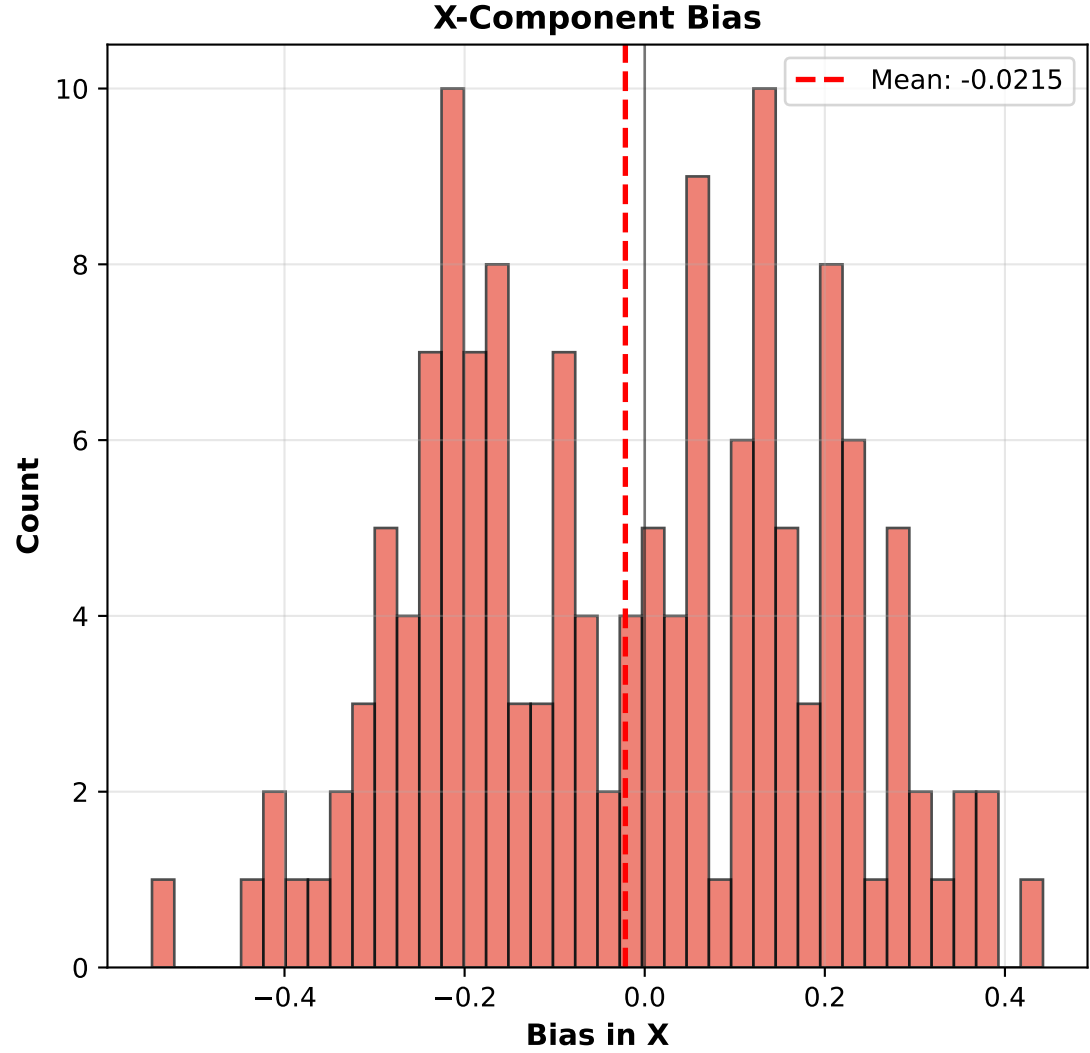
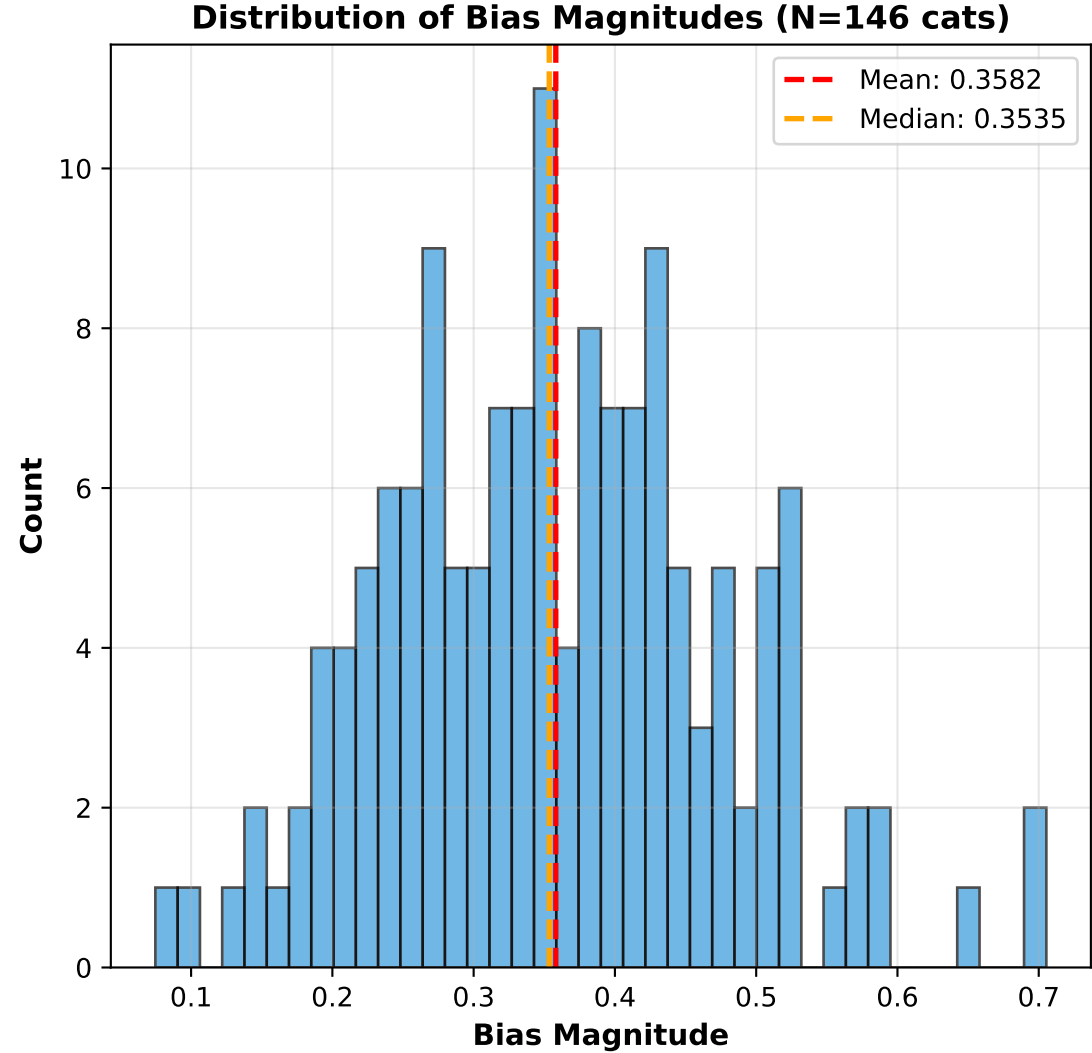
Neutrino Pointing Reconstruction: cat000039 - Perfect CT Scenario

Yellow: Electrons | Red: True Neutrino Direction

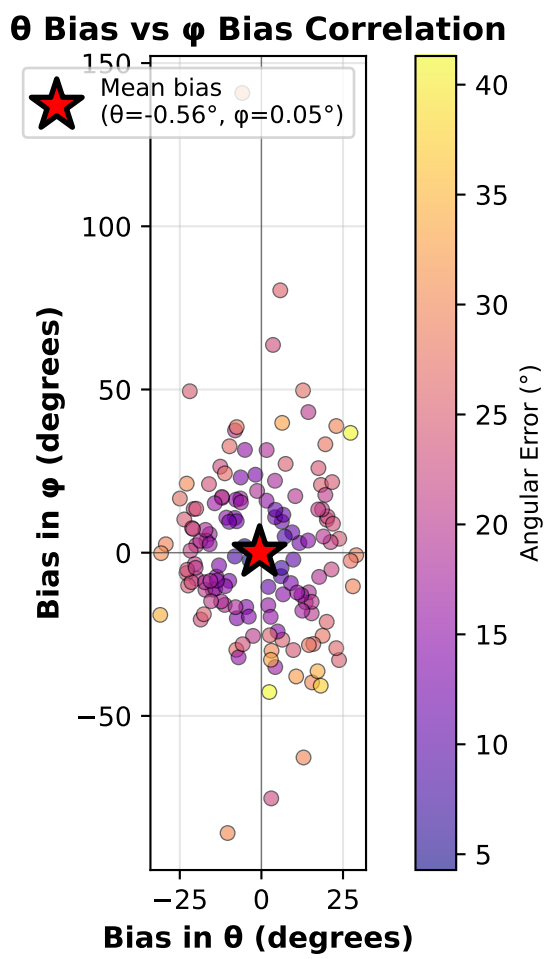
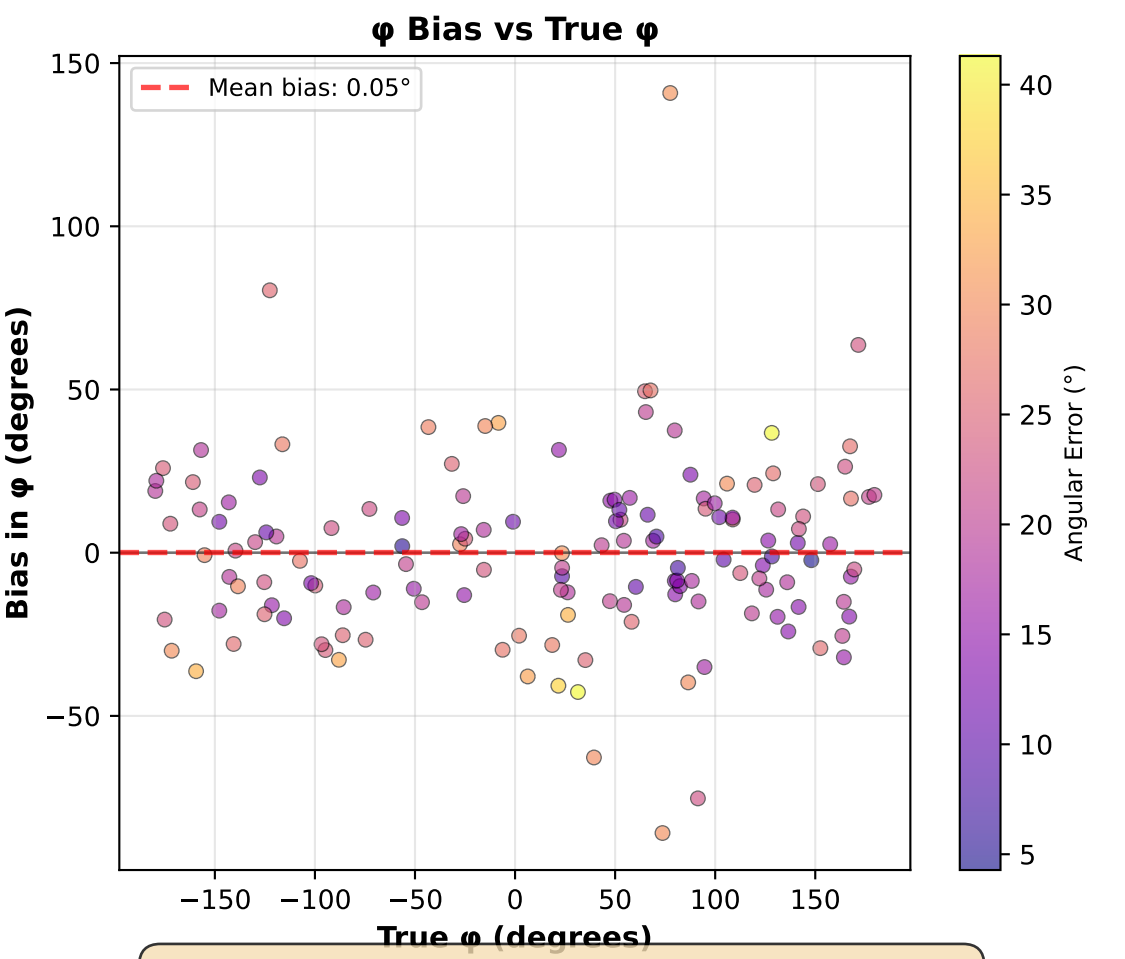
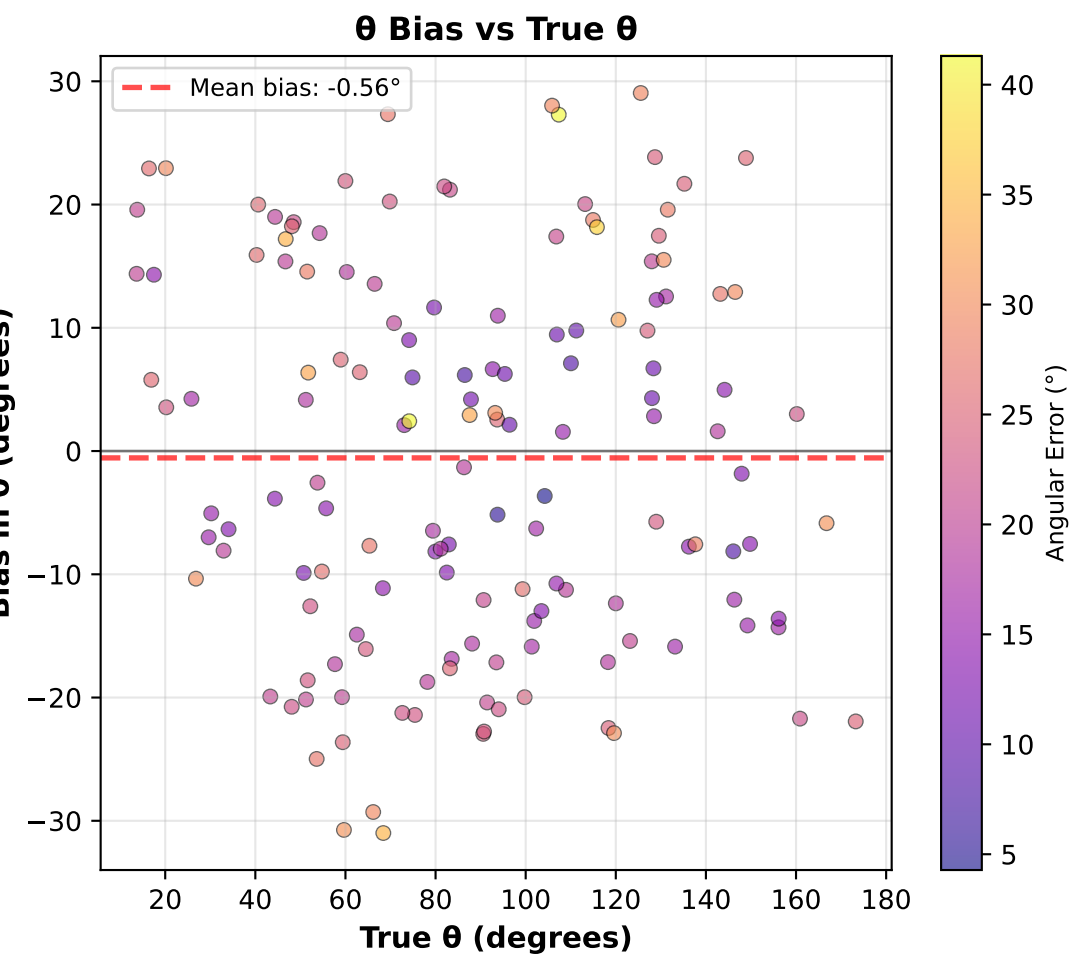
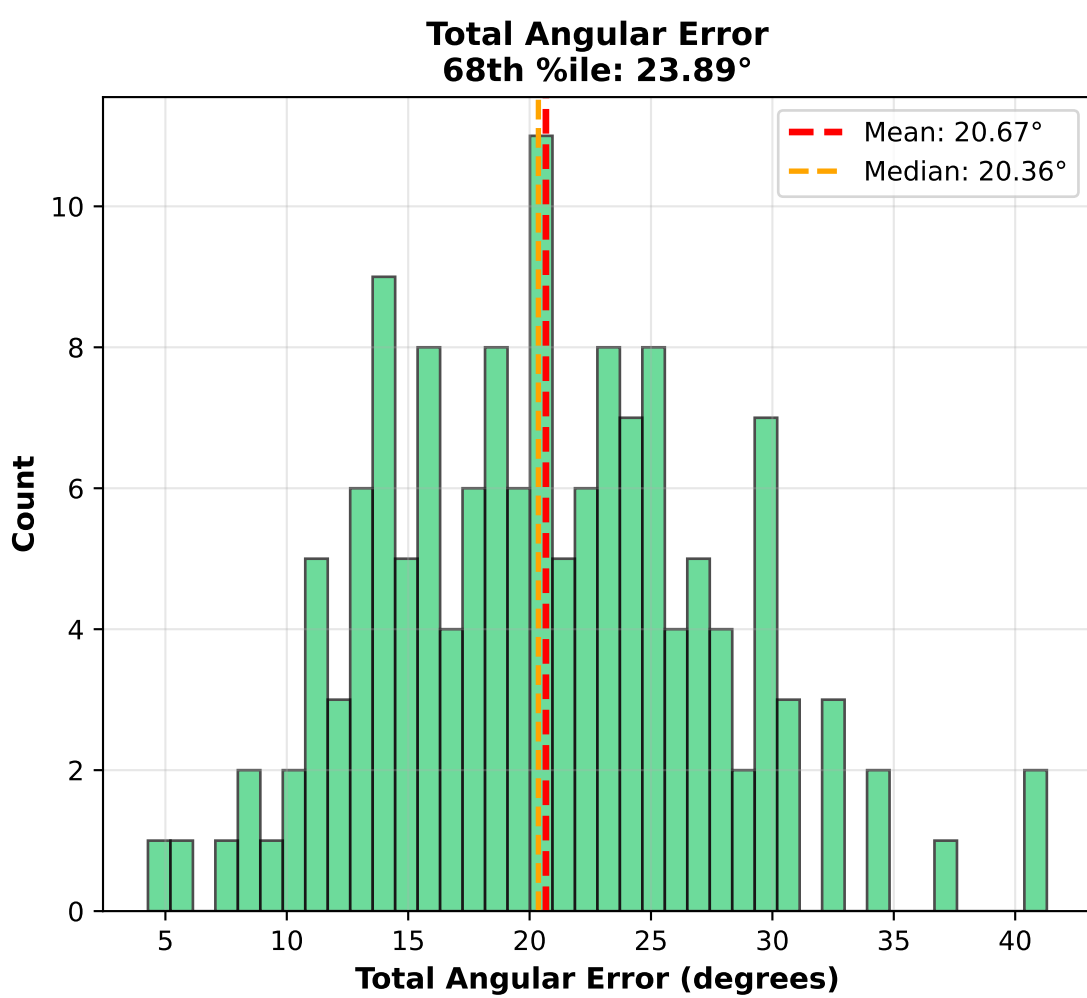
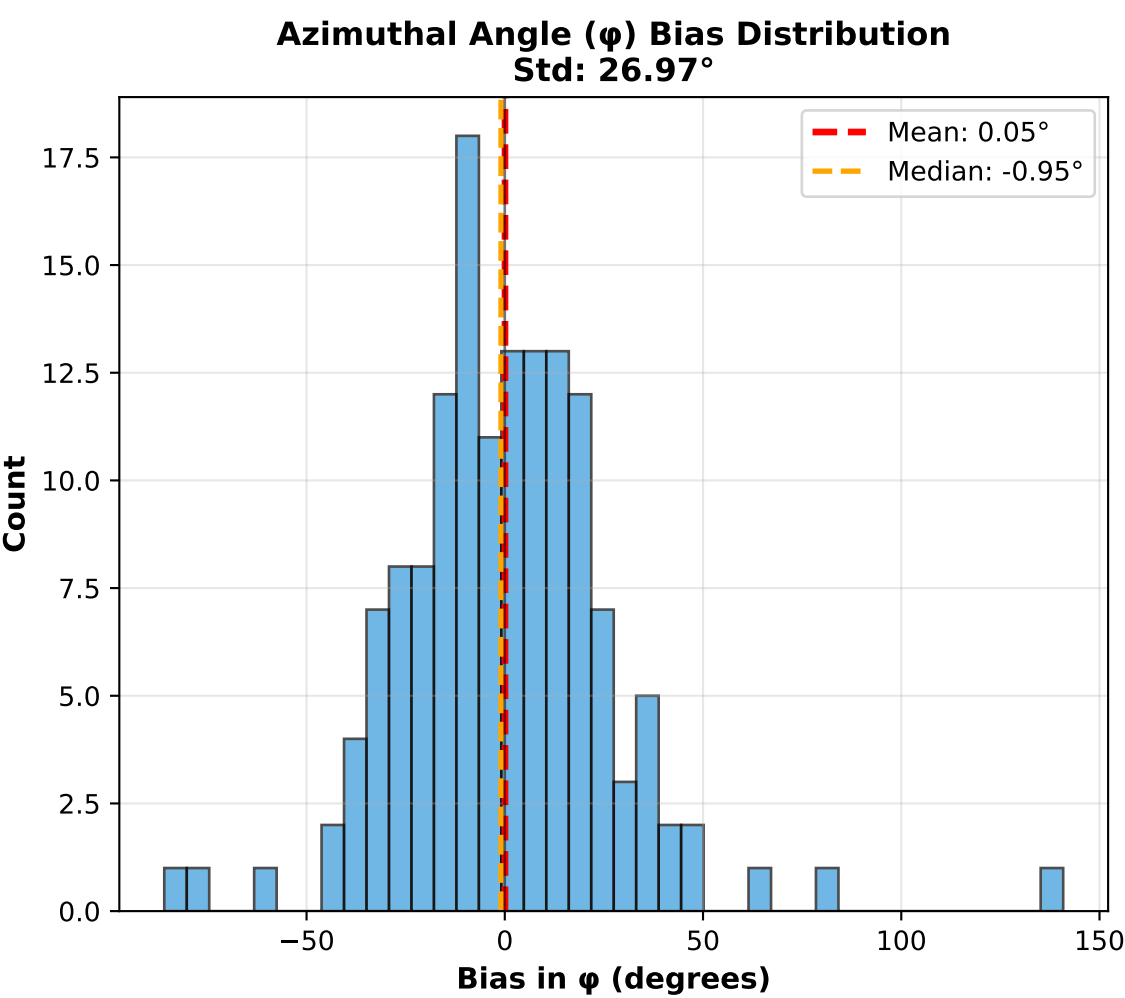
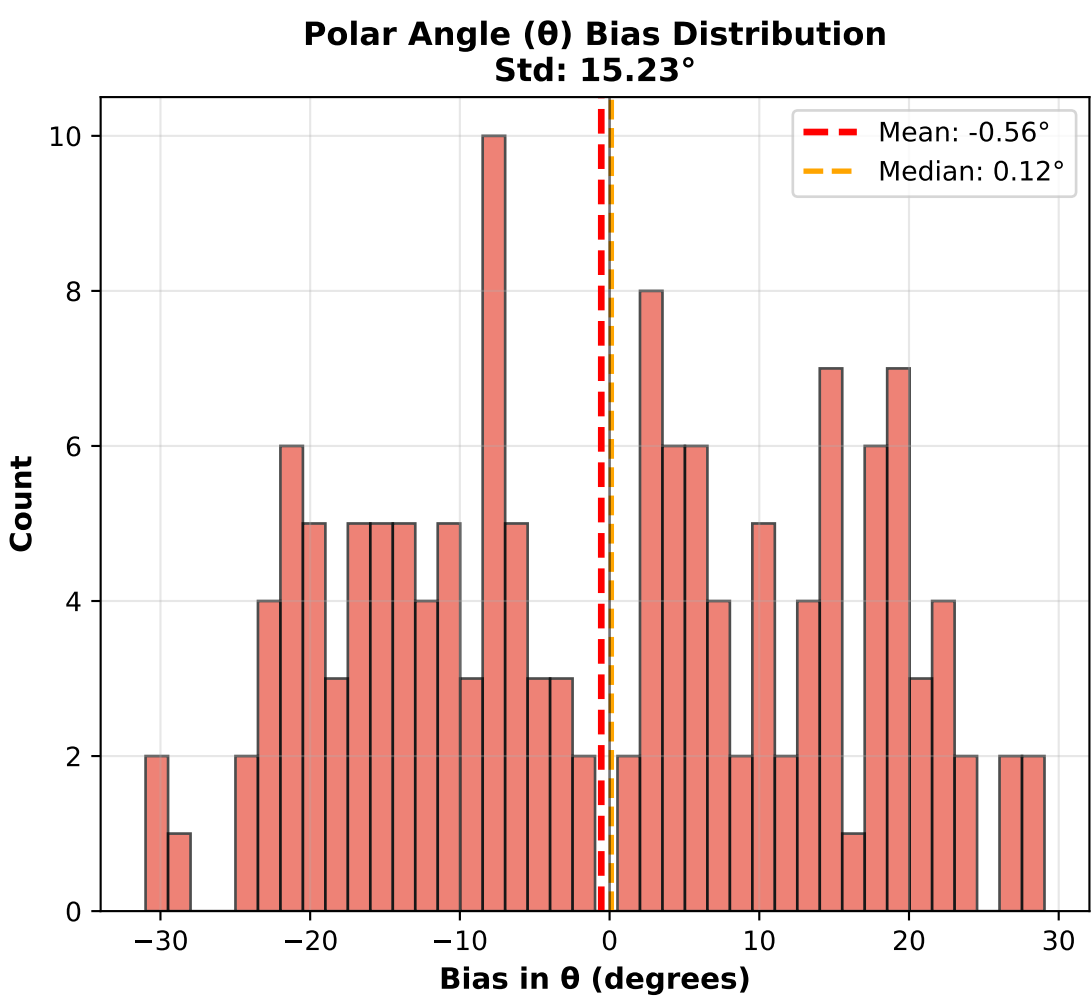


Best cat from EMCEE analysis: cat000039 | Angular Error: 0.39° | 68% Credible Interval (ω_{68}): 0.33° | N electrons: 200

Directional Bias Analysis (Perfect CT) - Cartesian Components
Mean Bias Vector: (-0.0215, -0.0285, 0.0119), Magnitude: 0.0377



Angular Bias Analysis in Spherical Coordinates (Perfect CT)
 θ = polar angle from z-axis [0°, 180°], ϕ = azimuthal angle from x-axis [-180°, 180°]



Angular Bias Statistics (N=146 cats):
θ bias: mean=-0.56°, median=0.12°, std=15.23°
φ bias: mean=0.05°, median=-0.95°, std=26.97°
Total angular error: mean=20.67°, 68th %ile=23.89°